10-701: Introduction to Machine Learning Lecture 7 – Logistic Regression

Henry Chai

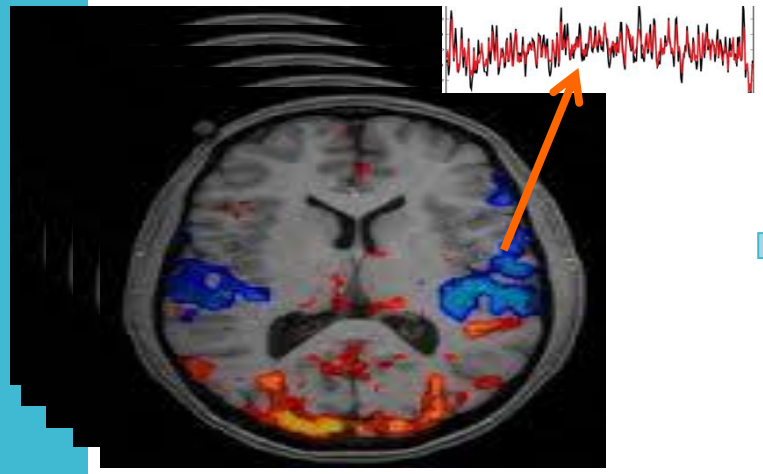
2/7/24

Front Matter

- Announcements:
  - HW2 released 2/7 (today!), due on 2/16 at 11:59 PM
- Recommended Readings:
  - Murphy, Section 8.1 - 8.3

Regression to Classification

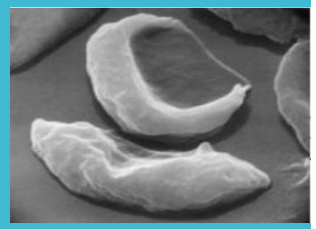
Regression

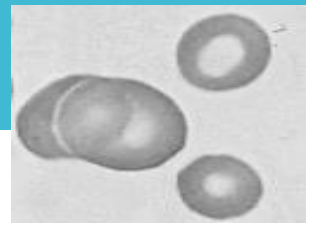


X = Brain Scan

Y = Age of a subject

Classification





X = Cell Image

Anemic cell
Healthy cell

Y = Diagnosis

Can we predict the "probability" of class label being Anemic or Healthy – a real number – using regression methods?

But output (probability) needs to be in [0,1]

Classification

Goal: Construct a **predictor** $f : X \to Y$ to minimize a risk (performance measure) $R(f)$

Sports
Science
News

Features, X

Labels, Y

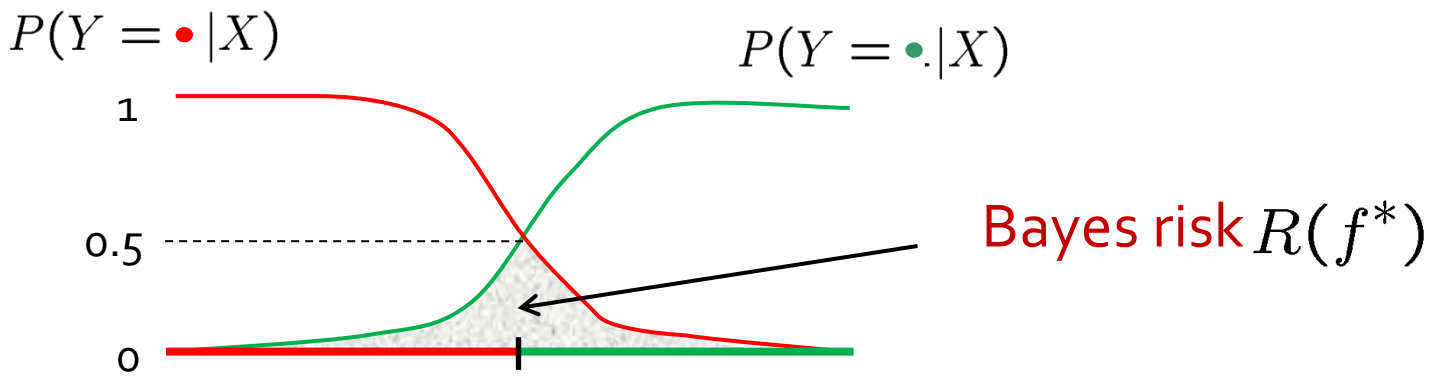
$$R(f) = P(f(X) \neq Y)$$

Probability of Error

Optimal Classification

Optimal predictor:
(Bayes classifier)

$$f^* = \arg\min_f P(f(X) \neq Y)$$



$$f^*(x) = \arg\max_{Y=y} P(Y = y|X = x)$$

- Even the optimal classifier makes mistakes R(f*) > 0
- Optimal classifier depends on **unknown** distribution P_{XY}

Optimal Classifier

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Bayes Rule:

$$P(Y = y|X = x) = \frac{P(X = x|Y = y)P(Y = y)}{P(X = x)}$$

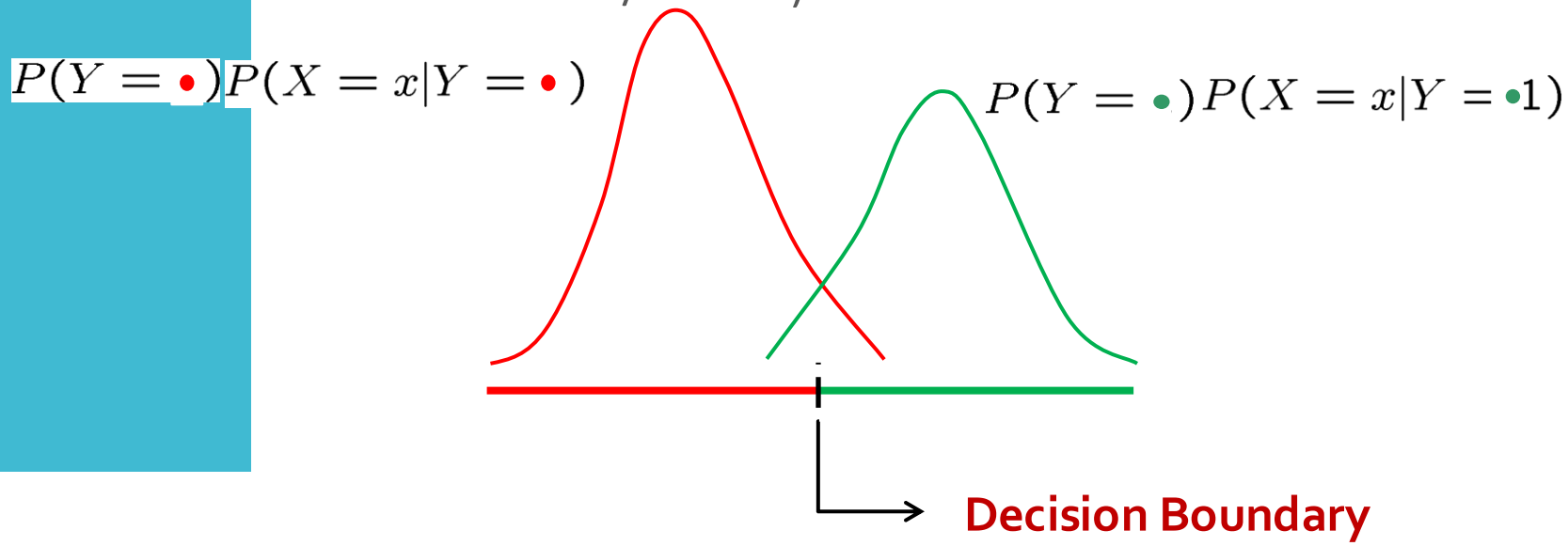
$$f^*(x) = \arg\max_{Y=y} P(Y = y|X = x)$$

$$= \arg\max_{Y=y} \underbrace{P(X = x|Y = y)}_{\text{Class conditional density}}\underbrace{P(Y = y)}_{\text{Class prior}}$$

Example Decision Boundaries

$$P(X = x|Y = y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x-\mu_y)^2}{2\sigma_y^2}\right)$$

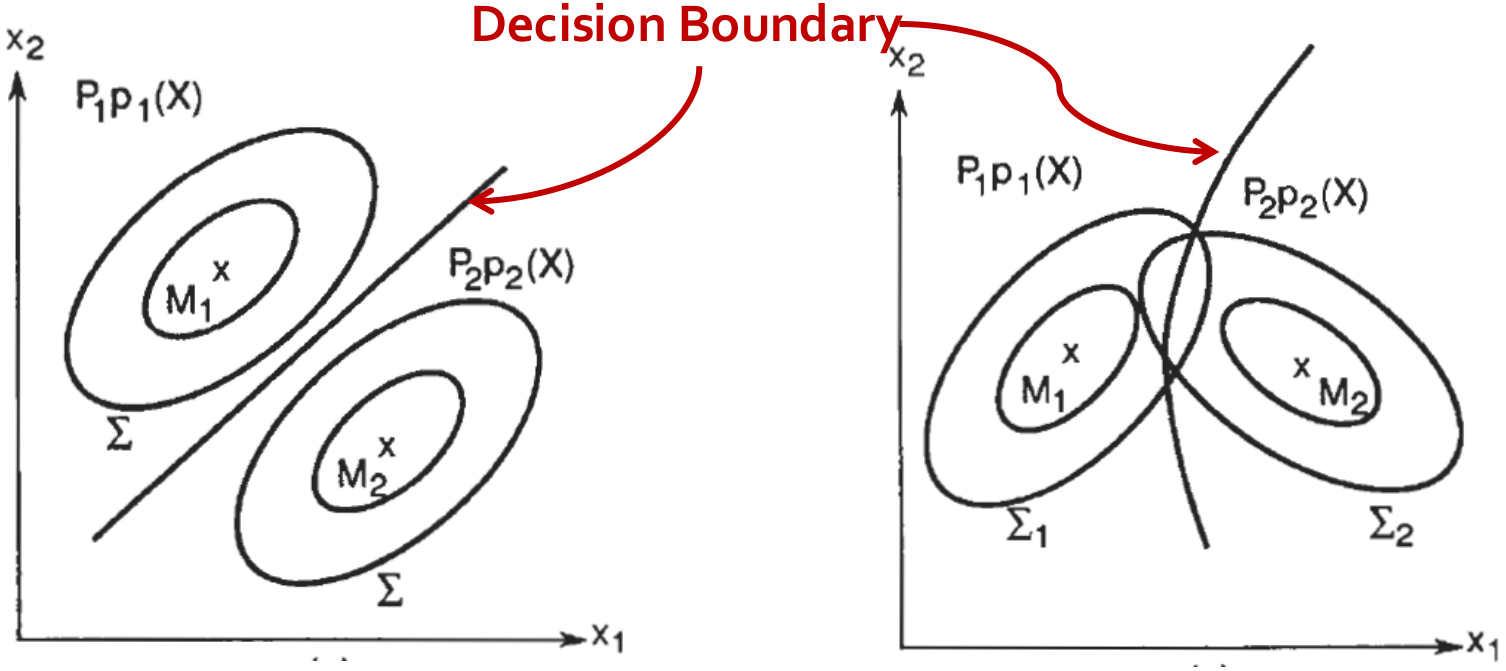
- Gaussian class conditional densities (1-dimension/feature)



Example Decision Boundaries

- Gaussian class conditional densities (2-dimensions/features)

$$P(X = x|Y = y) = \frac{1}{\sqrt{2\pi|\Sigma_y|}} \exp\left(-\frac{(x-\mu_y)\Sigma_y^{-1}(x-\mu_y)'}{2}\right)$$



Learning the Optimal Classifier

Optimal classifier:

$$f^*(x) = \arg\max_{Y=y} P(Y = y|X = x)$$

$$= \arg\max_{Y=y} \underbrace{P(X = x|Y = y)}_{\text{Class conditional density}} \underbrace{P(Y = y)}_{\text{Class prior}}$$

Need to know Prior P(Y = y) for all y

Likelihood P(X=x|Y = y) for all x,y

Generative vs. Discriminative Classifiers

Generative classifiers

- Assume some functional form for P(X,Y) (or P(X|Y) and P(Y))
- Estimate parameters of P(X|Y), P(Y) directly from training data
- Use Bayes rule to calculate P(Y|X)

Discriminative classifiers

- Assume some functional form for P(Y|X) or for the decision boundary
- Estimate parameters of P(Y|X) directly from training data

Probabilistic Classification

- Classification:
  - (Unknown) Target function, $c^*: \mathcal{X} \to \mathcal{Y}$
  - Classifier, $h : \mathcal{X} \to \mathcal{Y}$
  - Goal: find a classifier, h, that best approximates c^*
- Probabilistic Classification:
  - (Unknown) Target *distribution*, $y \sim P^*(Y|\boldsymbol{x})$
  - Distribution, $P(Y|\boldsymbol{x})$
  - Goal: find a distribution, P, that best approximates P^*

Building a Probabilistic Classifier

- Define a decision rule
  - Given a test data point $\boldsymbol{x}'$, predict its label $\hat{y}$ using the *posterior distribution* $P(Y = y | X = \boldsymbol{x}')$
  - Common choice: $\hat{y} = \underset{y}{\operatorname{argmax}}\, P(Y = y | X = \boldsymbol{x}')$
- Model the posterior distribution
  - Option 1 ("Discriminative Model") (today)
    - Model $P(Y|X)$ directly as some function of X
  - Option 2 ("Generative Model")
    - Use Bayes' rule:

$$P(Y|X) = \frac{P(X|Y)\, P(Y)}{P(X)} \propto P(X|Y)\, P(Y)$$

Modelling the Posterior

- Suppose we have binary labels $y \in \{0,1\}$ and D-dimensional inputs $\boldsymbol{x} = [1, x_1, \dots, x_D]^T \in \mathbb{R}^{D+1}$
- **Assume**

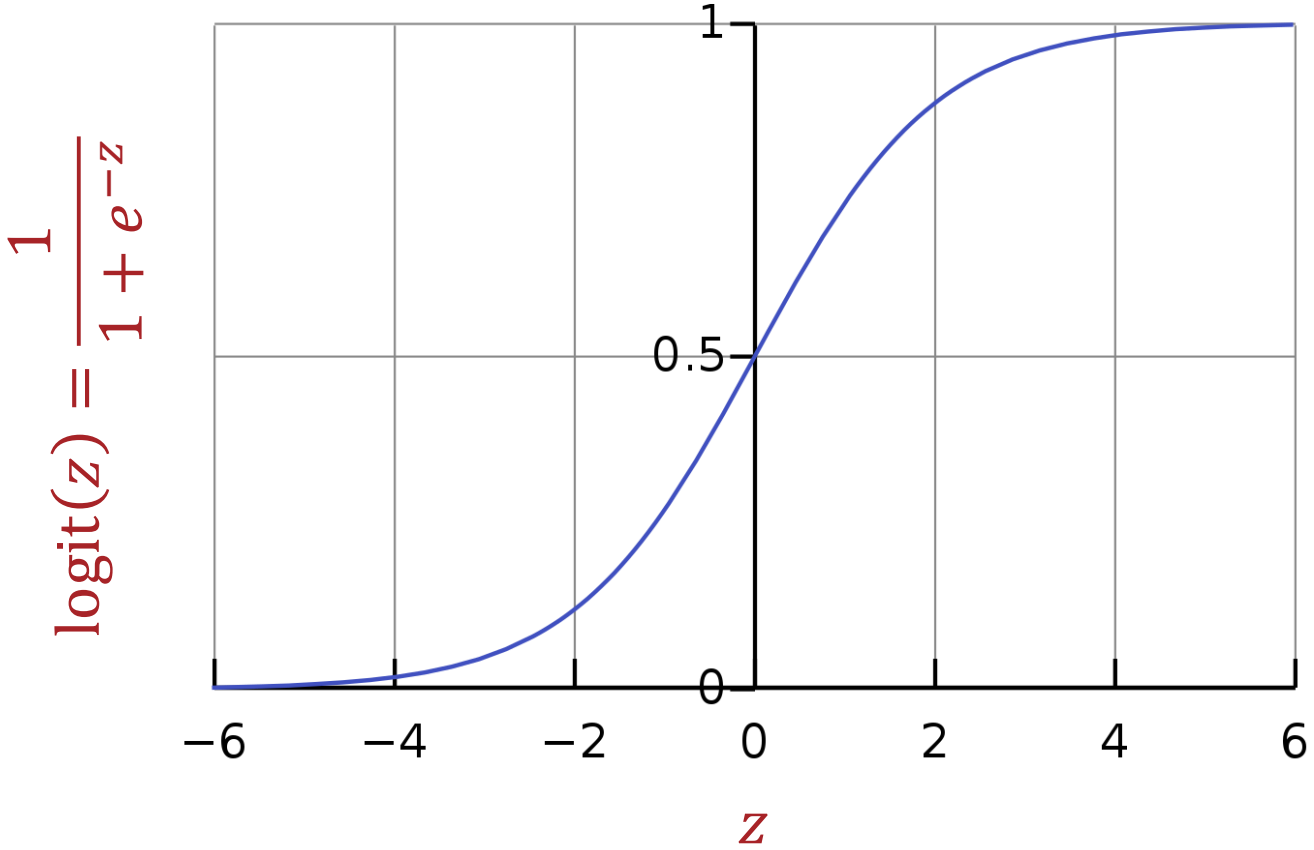
$$P(Y = 1|\boldsymbol{x}) = \text{logit}(\boldsymbol{w}^T\boldsymbol{x}) = \frac{1}{1 + \exp(-\boldsymbol{w}^T\boldsymbol{x})}$$

$$= \frac{\exp(\boldsymbol{w}^T\boldsymbol{x})}{\exp(\boldsymbol{w}^T\boldsymbol{x}) + 1}$$

- This implies two useful facts:

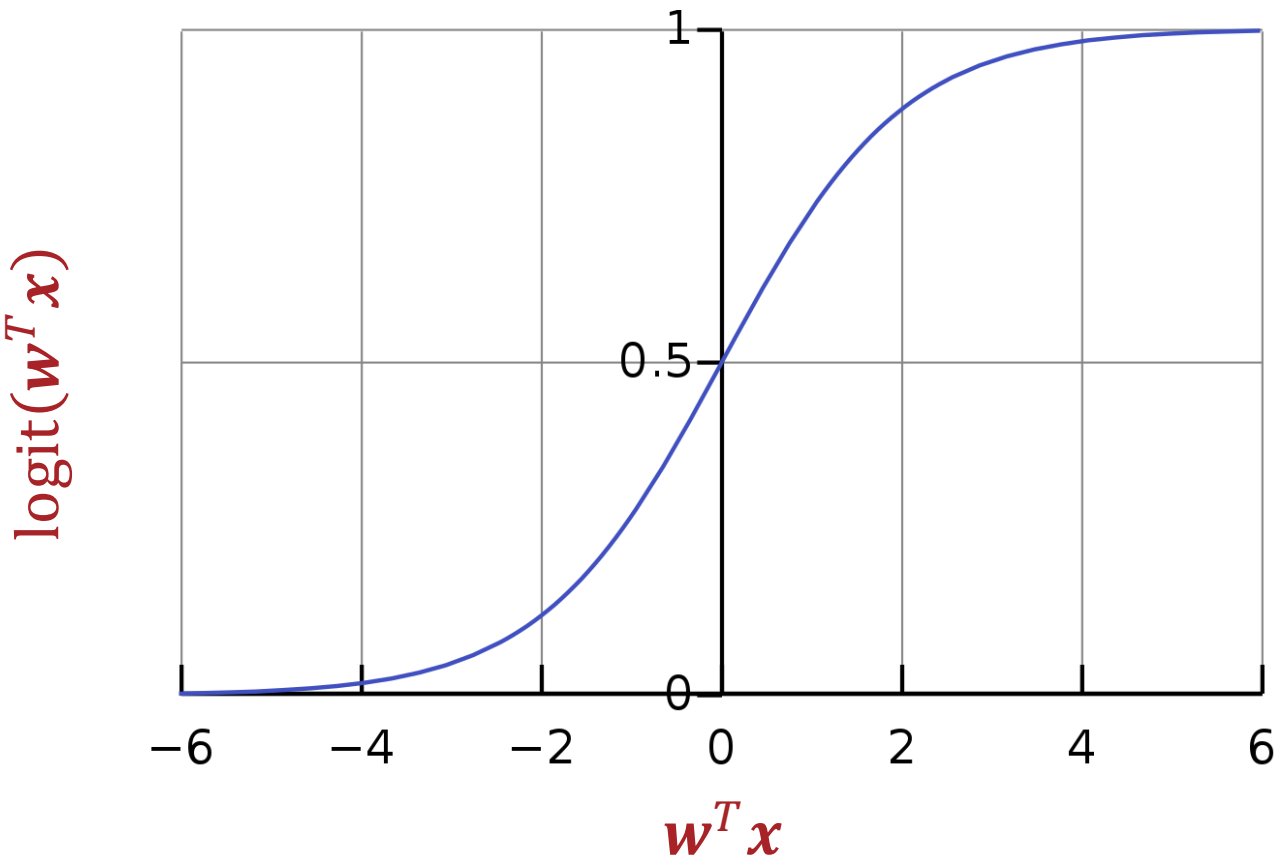
1. $P(Y = 0|\boldsymbol{x}) = 1 - P(Y = 1|\boldsymbol{x}) = \dfrac{1}{\exp(\boldsymbol{w}^T\boldsymbol{x}) + 1}$
2. $\dfrac{P(Y = 1|\boldsymbol{x})}{P(Y = 0|\boldsymbol{x})} = \exp(\boldsymbol{w}^T\boldsymbol{x}) \rightarrow \log\dfrac{P(Y = 1|\boldsymbol{x})}{P(Y = 0|\boldsymbol{x})} = \boldsymbol{w}^T\boldsymbol{x}$

Logistic Function



Source: https://en.wikipedia.org/wiki/Logistic_function#/media/File:Logistic-curve.svg

Why use the Logistic Function?



- Differentiable everywhere
- $\mathrm{logit}: \mathbb{R} \to [0, 1]$
- The decision boundary is linear in $\boldsymbol{x}$!

Source: https://en.wikipedia.org/wiki/Logistic_function#/media/File:Logistic-curve.svg

Logistic Regression Decision Boundary

$$\hat{y} = \begin{cases} 1 \text{ if } P(Y = 1|\boldsymbol{x}) \geq \frac{1}{2} \\ 0 \text{ otherwise} \end{cases}$$

$$P(Y = 1|\boldsymbol{x}) = \text{logit}(\boldsymbol{w}^T\boldsymbol{x}) = \frac{1}{1 + \exp(-\boldsymbol{w}^T\boldsymbol{x})} \geq \frac{1}{2}$$

$$2 \geq 1 + \exp(-\boldsymbol{w}^T\boldsymbol{x})$$

$$1 \geq \exp(-\boldsymbol{w}^T\boldsymbol{x})$$

$$\log(1) \geq -\boldsymbol{w}^T\boldsymbol{x}$$

$$0 \leq \boldsymbol{w}^T\boldsymbol{x}$$

Logistic Regression Decision Boundary

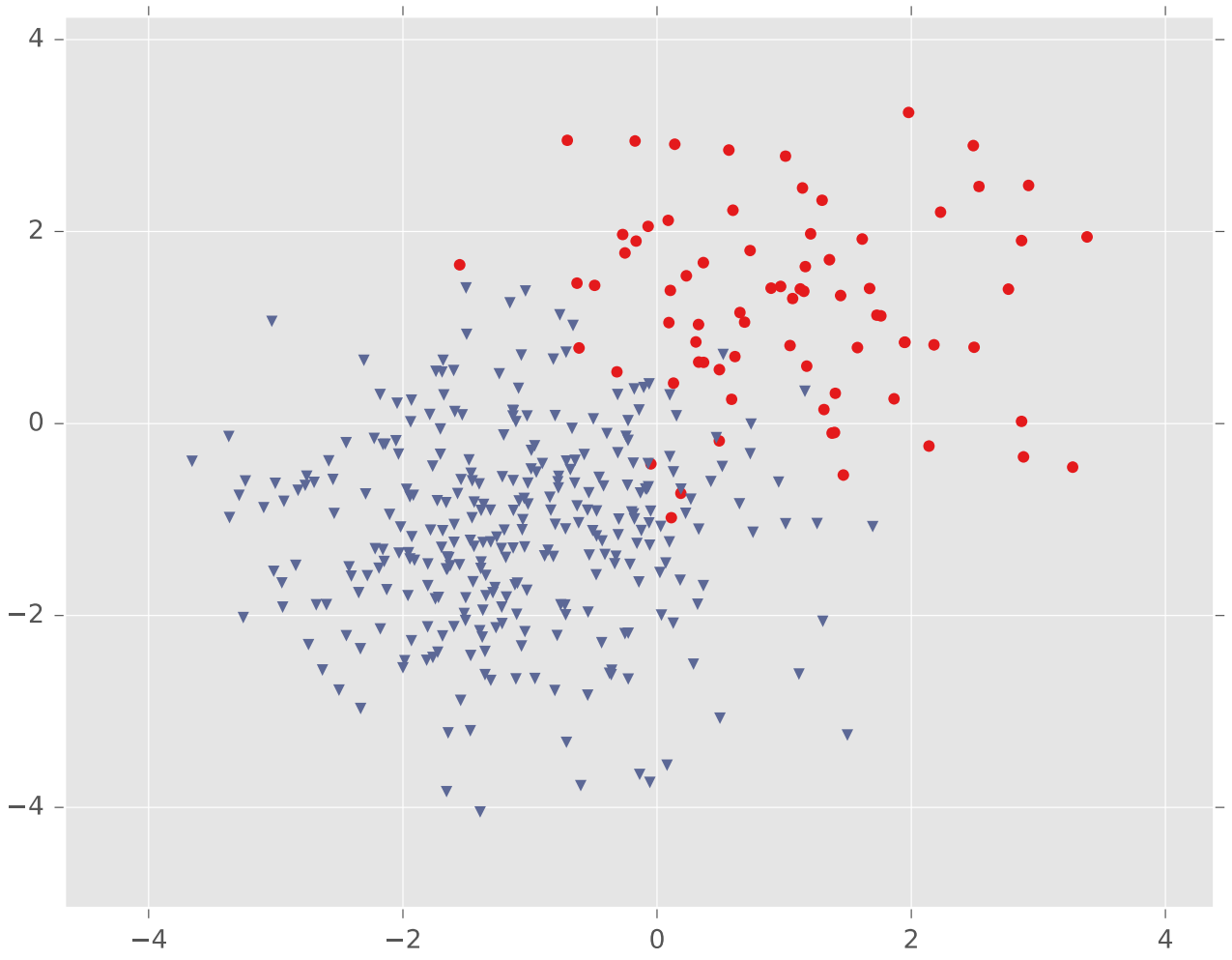


Figure courtesy of Matt Gormley

Logistic Regression Decision Boundary

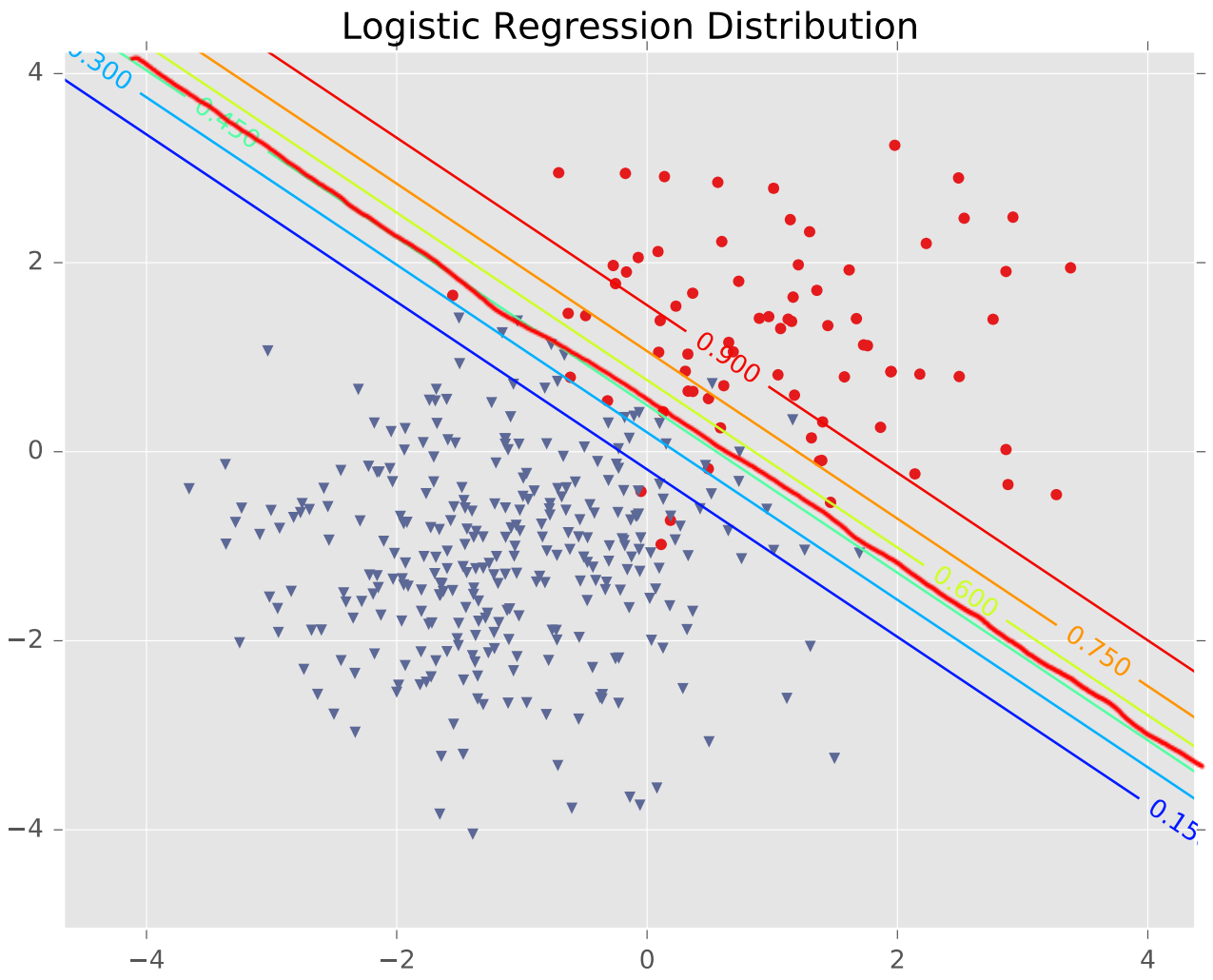


Figure courtesy of Matt Gormley

Logistic Regression Decision Boundary

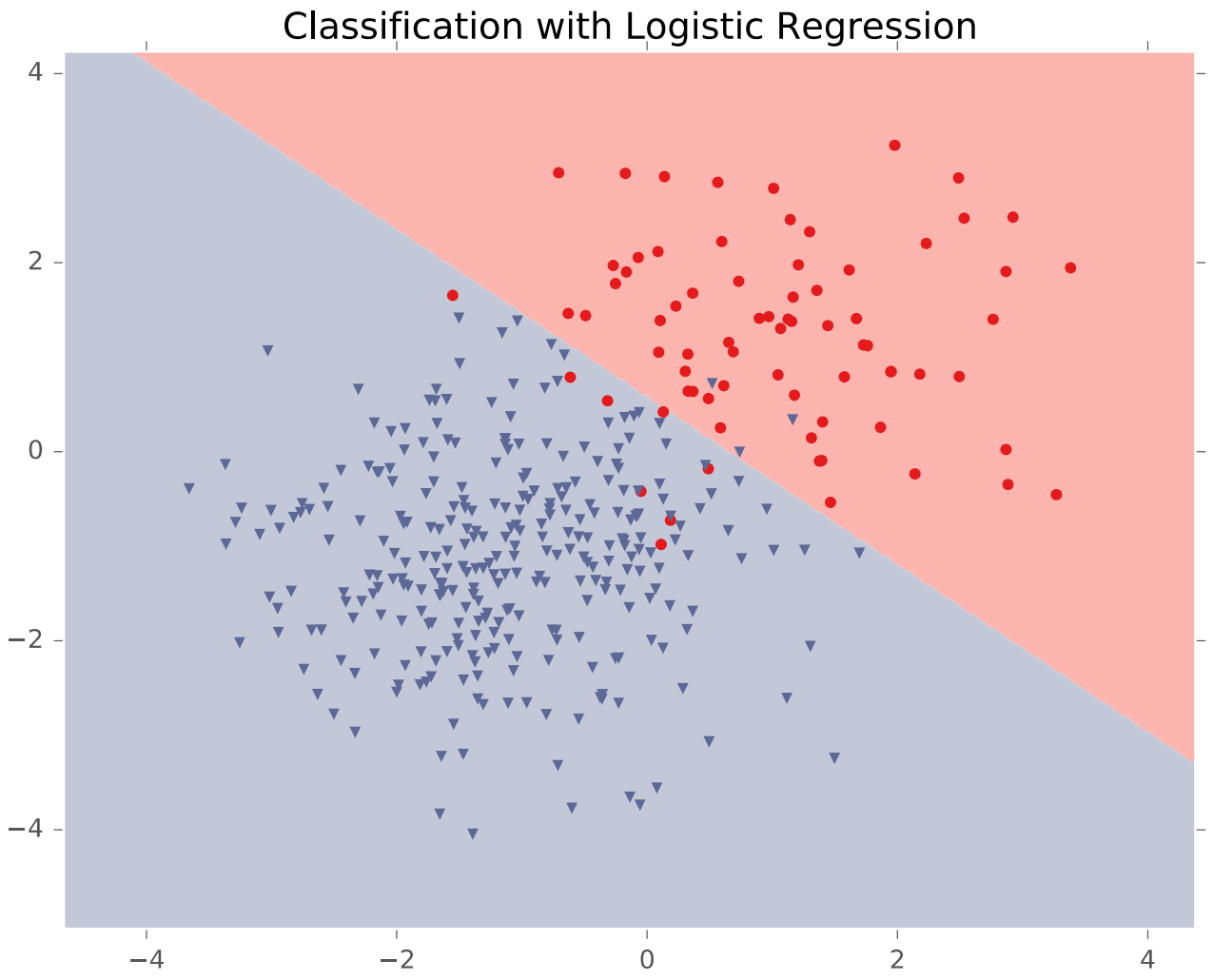


Figure courtesy of Matt Gormley

Logistic Regression

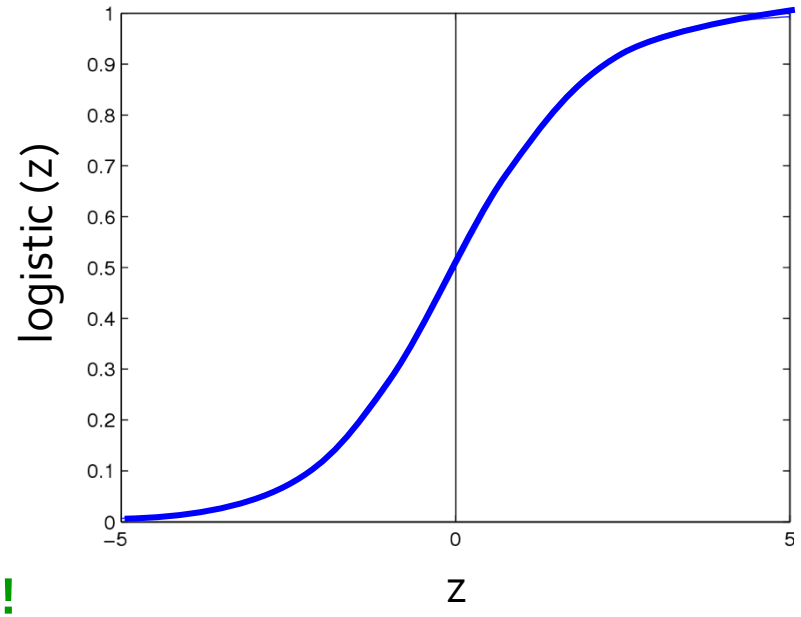
Not really regression

Assumes the following functional form for P(Y|X):

$$P(Y=0|X) = \frac{1}{1+\exp(w_0 + \sum_i w_i X_i)}$$

Logistic function applied to a linear function of the data

Logistic function (or Sigmoid): $\frac{1}{1+exp(-z)}$



Features can be discrete or continuous!

Logistic Regression is a Linear Classifier!

Assumes the following functional form for P(Y|X):

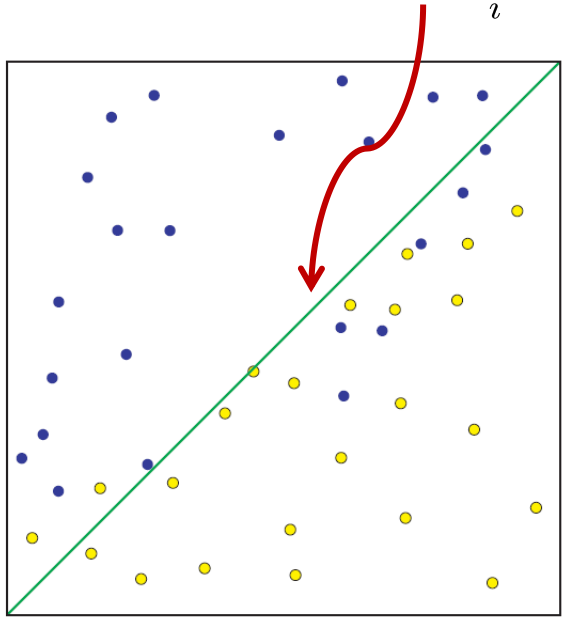
$$P(Y=0|X) = \frac{1}{1+\exp(w_0 + \sum_i w_i X_i)}$$

Decision boundary: Note - Labels are 0,1

$$P(Y=0|X) \underset{1}{\overset{0}{\gtrless}} P(Y=1|X)$$

$$w_0 + \sum_i w_i X_i \underset{0}{\overset{1}{\gtrless}} 0$$

$$w_0 + \sum_i w_i X_i = 0$$



(Linear Decision Boundary)

General Recipe for Machine Learning

- Define a model and model parameters
- Write down an objective function
- Optimize the objective w.r.t. the model parameters

Recipe for Logistic Regression

- Define a model and model parameters
  - Assume independent, identically distributed (iid) data
  - Assume $P(Y = 1|X) = \text{logit}(\boldsymbol{w}^T \boldsymbol{x})$
  - Parameters: $\boldsymbol{w} = [w_0, w_1, \dots, w_D]$
- Write down an objective function
  - ~~Maximize the *conditional* log-likelihood~~
  - Minimize the negative conditional log-likelihood
- Optimize the objective w.r.t. the model parameters
  - ???

Setting the Parameters via Minimum Negative Conditional (log-)Likelihood Estimation (MCLE)

Find $\boldsymbol{w}$ that minimizes

$$\ell_{\mathcal{D}}(\boldsymbol{w}) = -\log P\left(y^{(1)}, \dots, y^{(N)} \middle| \boldsymbol{x}^{(1)}, \dots, \boldsymbol{x}^{(N)}, \boldsymbol{w}\right) = -\log \prod_{n=1}^{N} P\left(y^{(n)} \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right)$$

$$= -\log \prod_{n=1}^{N} P\left(Y = 1 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right)^{y^{(n)}} \left(P\left(Y = 0 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right)\right)^{1-y^{(n)}}$$

$$= -\sum_{i=1}^{N} y^{(n)} \log P\left(Y = 1 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right) + \left(1 - y^{(n)}\right) \log P\left(Y = 0 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right)$$

$$= -\sum_{i=1}^{N} y^{(n)} \log \frac{P\left(Y = 1 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right)}{P\left(Y = 0 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right)} + \log P\left(Y = 0 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right)$$

$$= -\sum_{i=1}^{N} y^{(n)} \boldsymbol{w}^T \boldsymbol{x}^{(n)} - \log\left(1 + \exp\left(\boldsymbol{w}^T \boldsymbol{x}^{(n)}\right)\right)$$

Minimizing the Negative Conditional (log-)Likelihood

$$\ell_{\mathcal{D}}(\boldsymbol{w}) = -\sum_{n=1}^{N} y^{(n)} \boldsymbol{w}^T \boldsymbol{x}^{(n)} - \log\left(1 + \exp\left(\boldsymbol{w}^T \boldsymbol{x}^{(n)}\right)\right)$$

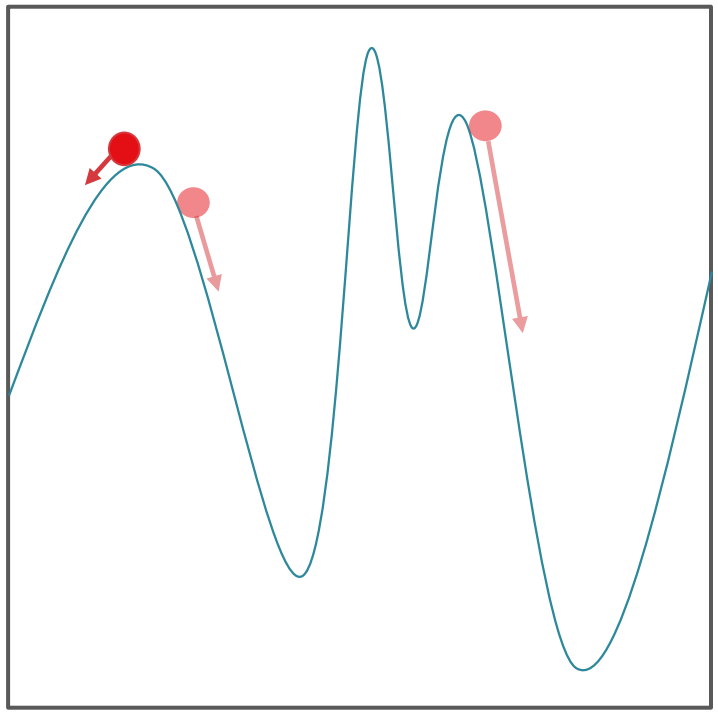
$$\nabla_{\boldsymbol{w}} \ell_{\mathcal{D}}(\boldsymbol{w}) = -\sum_{n=1}^{N} y^{(n)} \nabla_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{x}^{(n)} - \nabla_{\boldsymbol{w}} \log\left(1 + \exp\left(\boldsymbol{w}^T \boldsymbol{x}^{(n)}\right)\right)$$

$$= -\sum_{n=1}^{N} y^{(n)} \boldsymbol{x}^{(n)} - \frac{\exp\left(\boldsymbol{w}^T \boldsymbol{x}^{(n)}\right)}{1 + \exp\left(\boldsymbol{w}^T \boldsymbol{x}^{(n)}\right)} \boldsymbol{x}^{(n)}$$

$$= \sum_{n=1}^{N} \boldsymbol{x}^{(n)} \left(P\left(Y = 1 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}\right) - y^{(n)}\right)$$

Recall: Gradient Descent

- An iterative method for minimizing functions
- Requires the gradient to exist everywhere



Recall: Gradient Descent

- An iterative method for minimizing functions
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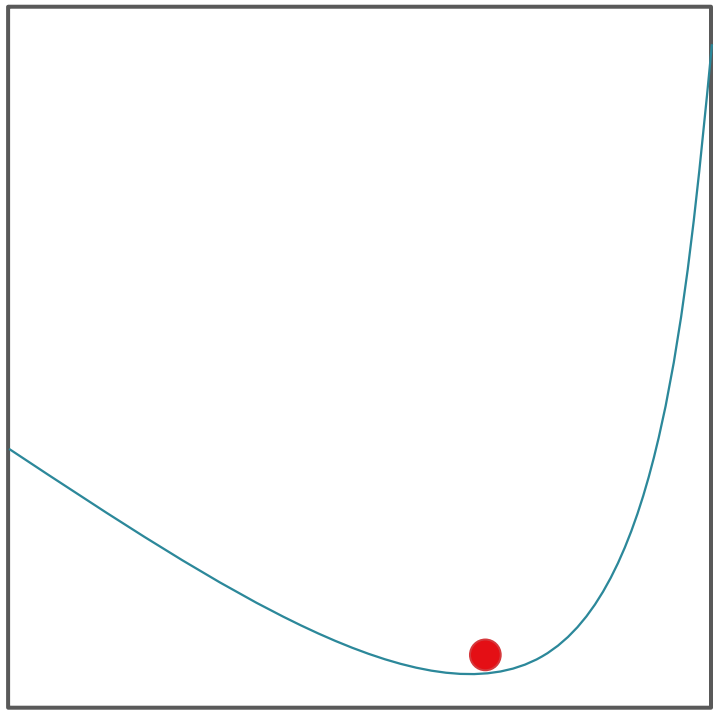


- Good news: the negative conditional log-likelihood, like the squared error, is also convex!

Gradient Descent

- Input: $\mathcal{D} = \left\{\left(\boldsymbol{x}^{(n)}, y^{(n)}\right)\right\}_{n=1}^{N}, \eta^{(0)}$

1. Initialize $\boldsymbol{w}^{(0)}$ to all zeros and set $t = 0$
2. While TERMINATION CRITERION is not satisfied
   a. Compute the gradient:

$$O(ND)\left\{\nabla_{\boldsymbol{w}} \ell_{\mathcal{D}}\left(\boldsymbol{w}^{(t)}\right) = \sum_{n=1}^{N} \boldsymbol{x}^{(n)}\left(P\left(Y = 1 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}^{(t)}\right) - y^{(n)}\right)\right.$$

   b. Update $\boldsymbol{w}$: $\boldsymbol{w}^{(t+1)} \leftarrow \boldsymbol{w}^{(t)} - \eta^{(0)} \nabla_{\boldsymbol{w}} \ell_{\mathcal{D}}\left(\boldsymbol{w}^{(t)}\right)$
   c. Increment t: $t \leftarrow t + 1$

- Output: $\boldsymbol{w}^{(t)}$

Stochastic Gradient Descent

- Input: $\mathcal{D} = \left\{\left(\boldsymbol{x}^{(n)}, y^{(n)}\right)\right\}_{n=1}^{N}, \eta_{SGD}^{(0)}$

1. Initialize $\boldsymbol{w}^{(0)}$ to all zeros and set $t = 0$
2. While TERMINATION CRITERION is not satisfied
   a. Randomly sample a data point from $\mathcal{D}$, $\left(\boldsymbol{x}^{(n)}, y^{(n)}\right)$
   b. Compute the pointwise gradient:

      $$\nabla_{\boldsymbol{w}} \ell^{(n)}\left(\boldsymbol{w}^{(t)}\right) = \boldsymbol{x}^{(n)}\left(P\left(Y = 1 \middle| \boldsymbol{x}^{(n)}, \boldsymbol{w}^{(t)}\right) - y^{(n)}\right)$$

   c. Update $\boldsymbol{w}$: $\boldsymbol{w}^{(t+1)} \leftarrow \boldsymbol{w}^{(t)} - \eta_{SGD}^{(0)} \nabla_{\boldsymbol{w}} \ell^{(n)}\left(\boldsymbol{w}^{(t)}\right)$
   d. Increment t: $t \leftarrow t + 1$

- Output: $\boldsymbol{w}^{(t)}$

Stochastic Gradient Descent

- If the data point is sampled uniformly at random, then the expected value of the pointwise gradient is proportional to the full gradient:

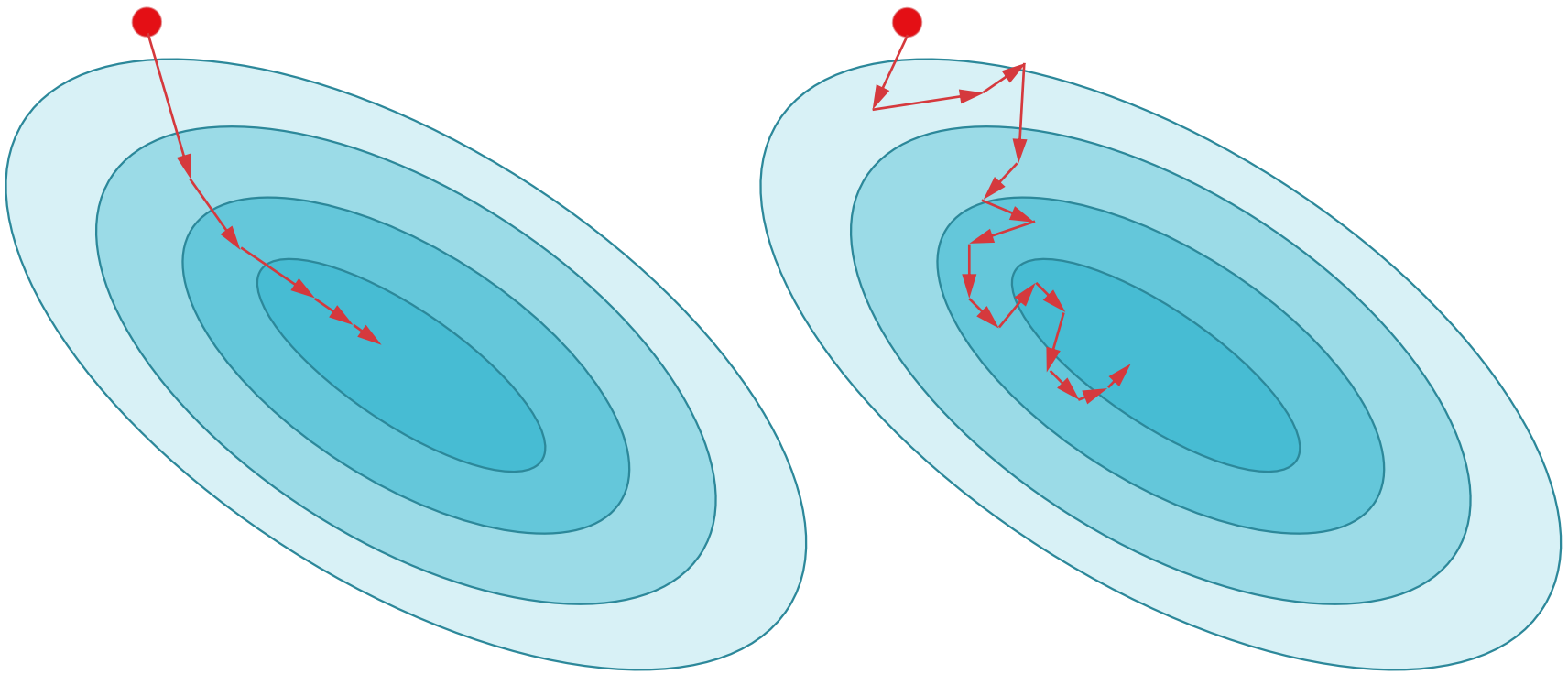
$$E\left[\nabla_{\boldsymbol{w}}\ell_{\boldsymbol{x}^{(n)},y^{(n)}}\left(\boldsymbol{w}^{(t)}\right)\right] = \frac{1}{N}\sum_{n=1}^{N}\nabla_{\boldsymbol{w}}\ell^{(n)}\left(\boldsymbol{w}^{(t)}\right)$$

$$= \frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}^{(n)}\left(P\left(Y=1\middle|\boldsymbol{x}^{(n)},\boldsymbol{w}^{(t)}\right)-y^{(n)}\right)$$

$$= \frac{1}{N}\nabla_{\boldsymbol{w}}\ell_{\mathcal{D}}\left(\boldsymbol{w}^{(t)}\right)$$

- In practice, the data set is randomly shuffled then looped through so that each data point is used equally often

Stochastic Gradient Descent vs. Gradient Descent



Gradient Descent

Stochastic Gradient Descent

Mini-batch Stochastic Gradient Descent

- Input: $\mathcal{D} = \{(\boldsymbol{x}^{(n)}, y^{(n)})\}_{n=1}^{N}, \eta_{MB}^{(0)}, B$

1. Initialize $\boldsymbol{w}^{(0)}$ to all zeros and set $t = 0$
2. While TERMINATION CRITERION is not satisfied
   a. Randomly sample B data points from $\mathcal{D}$:

$$\mathcal{D}_{batch}\{(\boldsymbol{x}^{(b)}, y^{(b)})\}_{b=1}^{B}$$

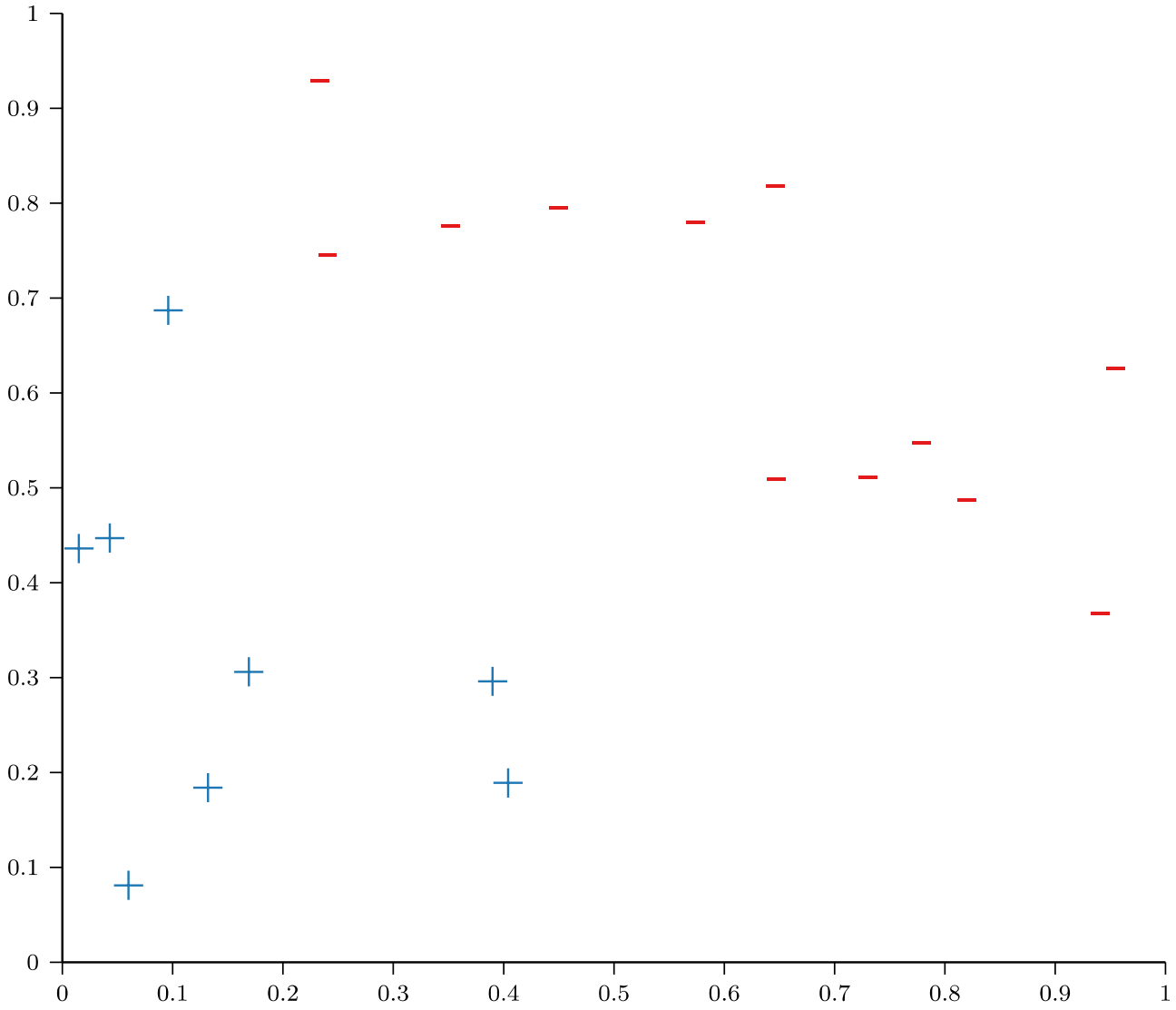
   b. Compute the gradient w.r.t. the sampled *batch*:

$$\nabla_{\boldsymbol{w}} \ell_{\mathcal{D}_{batch}}(\boldsymbol{w}^{(t)}) = \sum_{b=1}^{B} \boldsymbol{x}^{(b)}\left(P\left(Y = 1 \middle| \boldsymbol{x}^{(b)}, \boldsymbol{w}\right) - y^{(b)}\right)$$

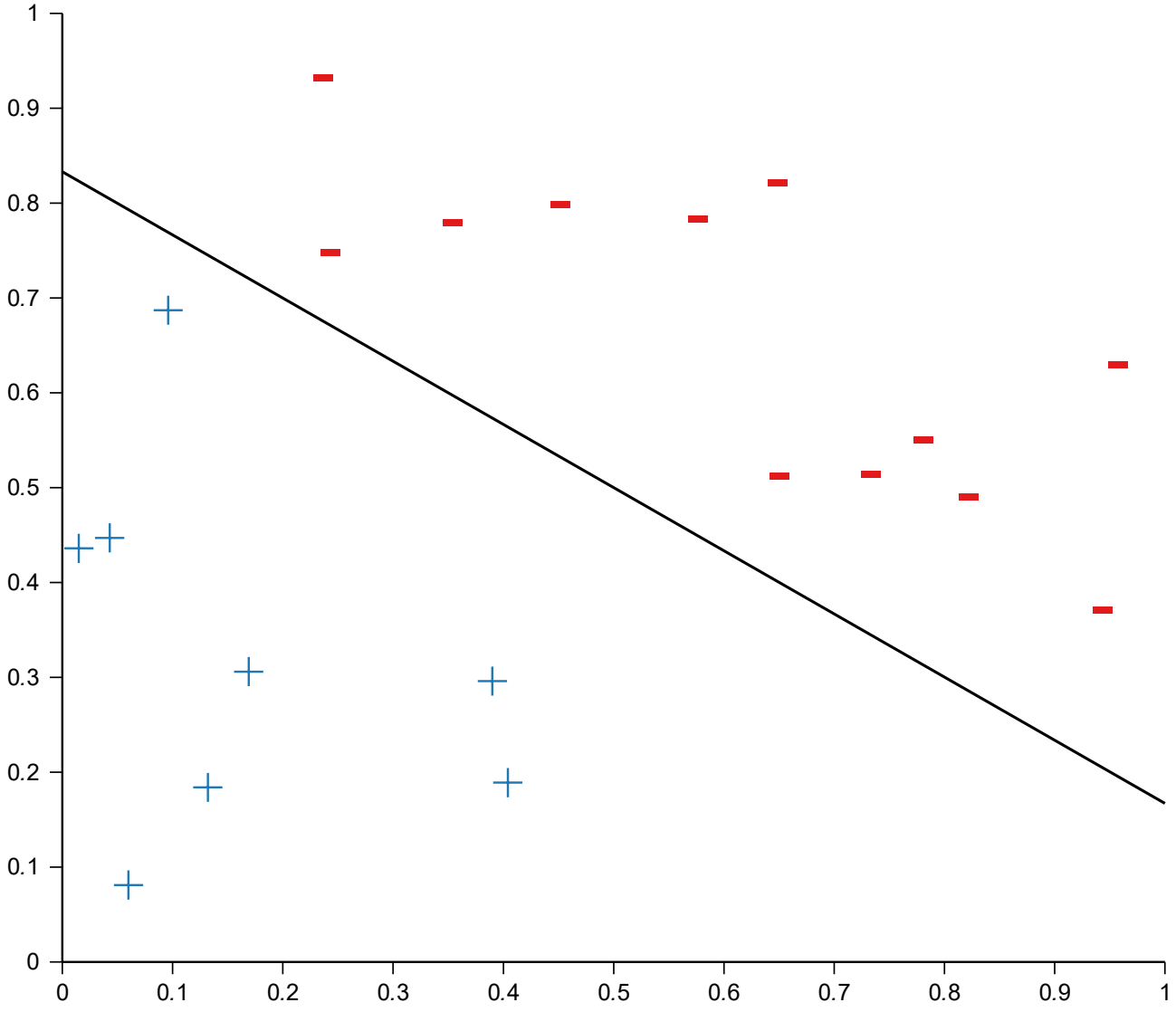
   c. Update $\boldsymbol{w}$: $\boldsymbol{w}^{(t+1)} \leftarrow \boldsymbol{w}^{(t)} - \eta_{MB}^{(0)} \nabla_{\boldsymbol{w}} \ell_{\mathcal{D}_{batch}}(\boldsymbol{w}^{(t)})$
   d. Increment t: $t \leftarrow t + 1$

- Output: $\boldsymbol{w}^{(t)}$

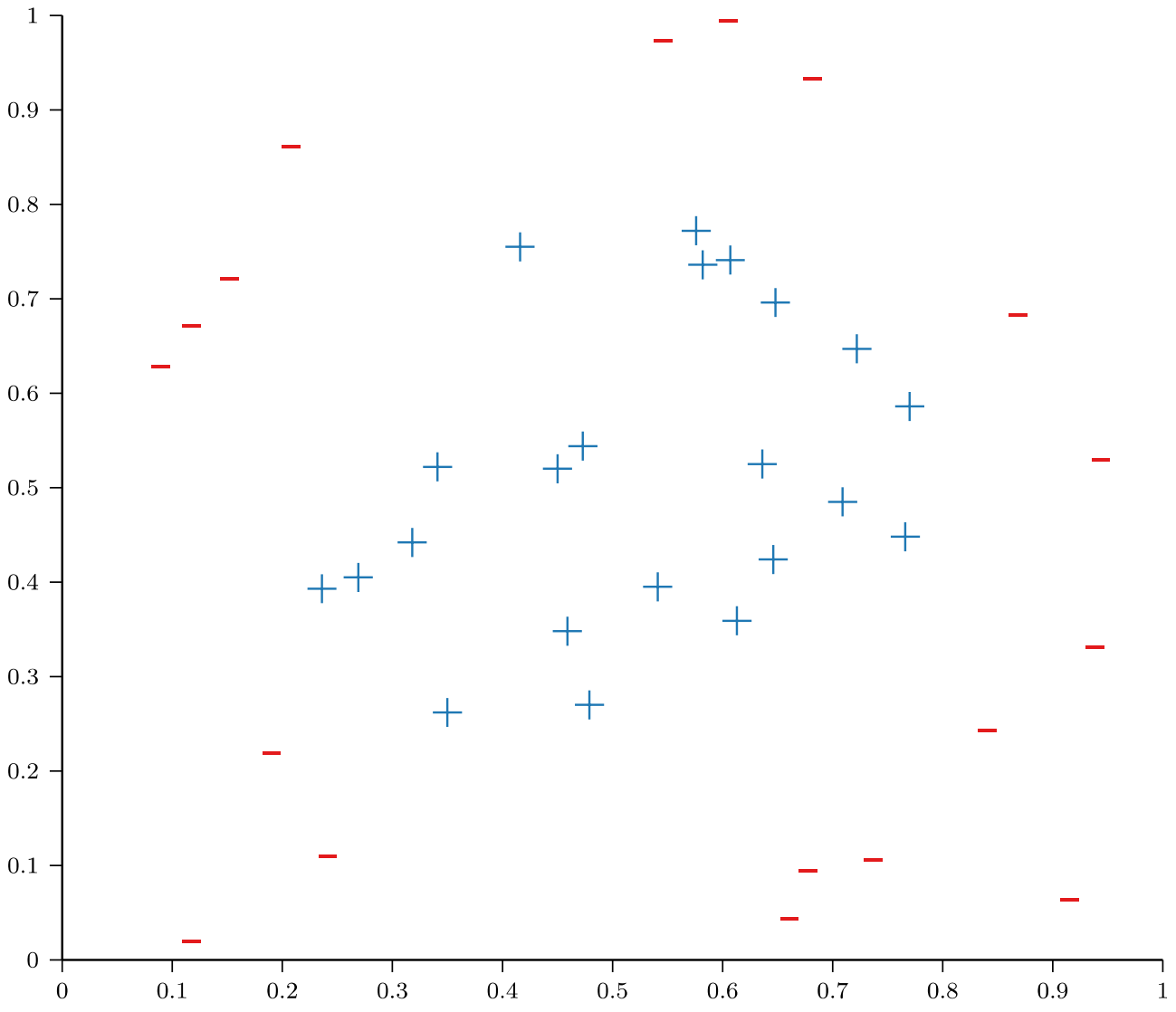
Linear Models



Linear Models



Linear Models?



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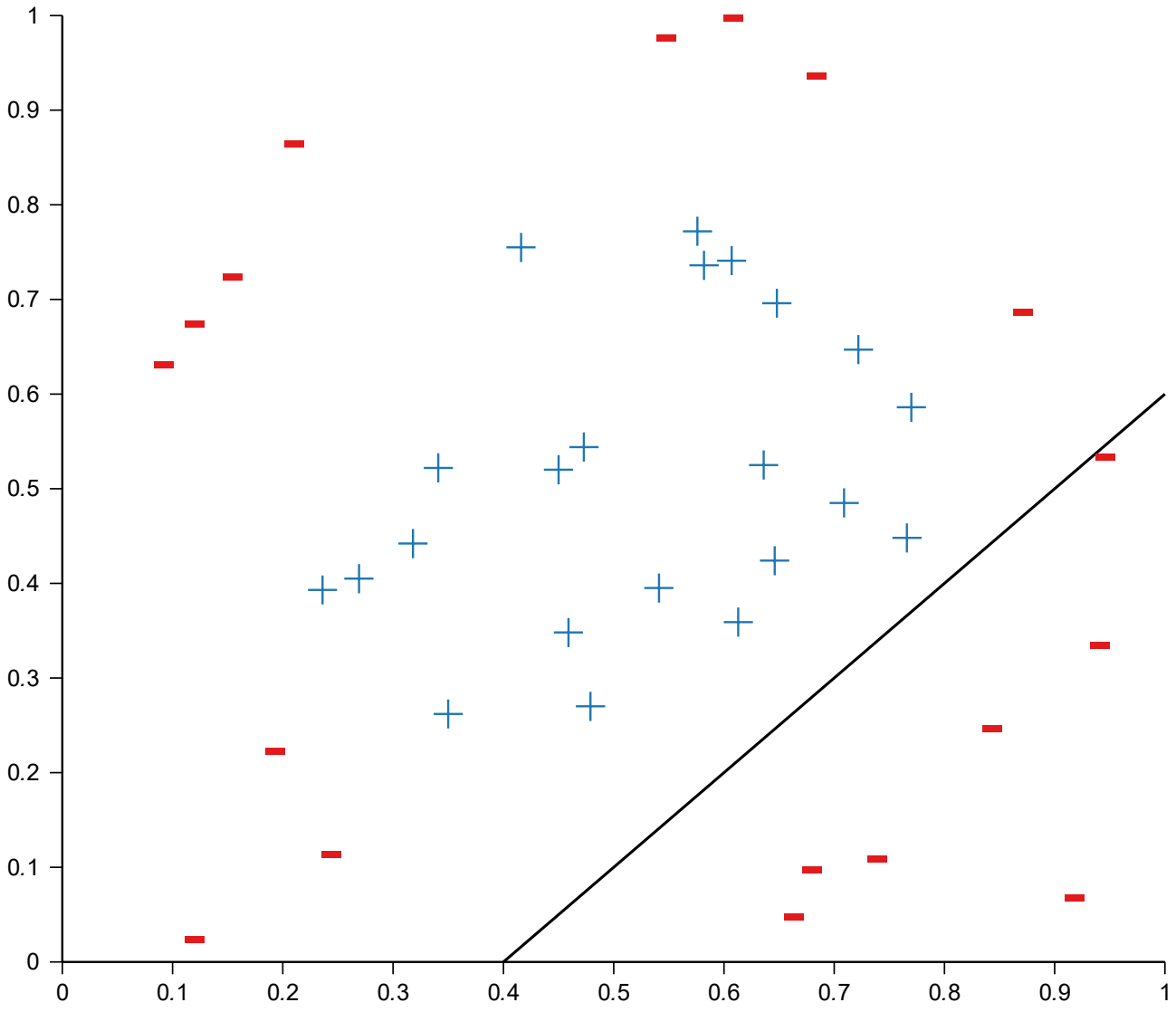
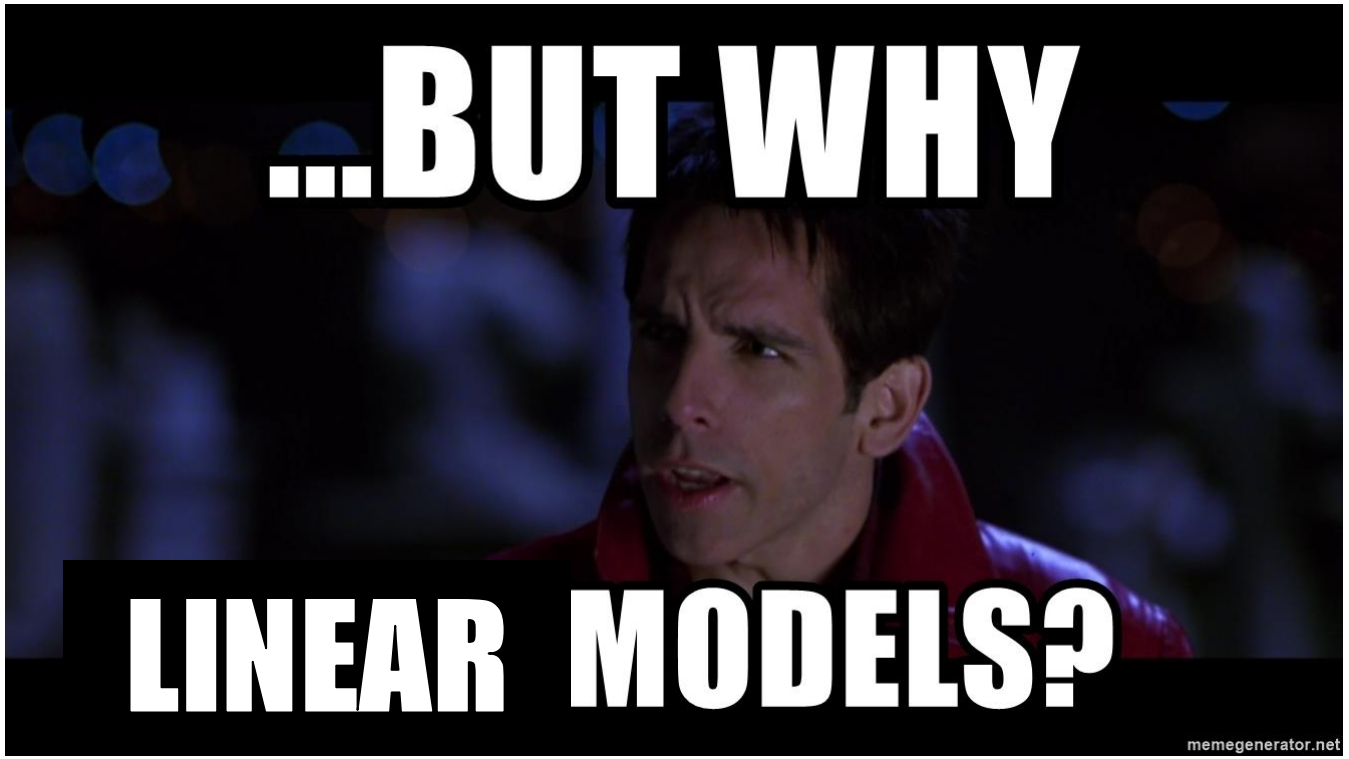
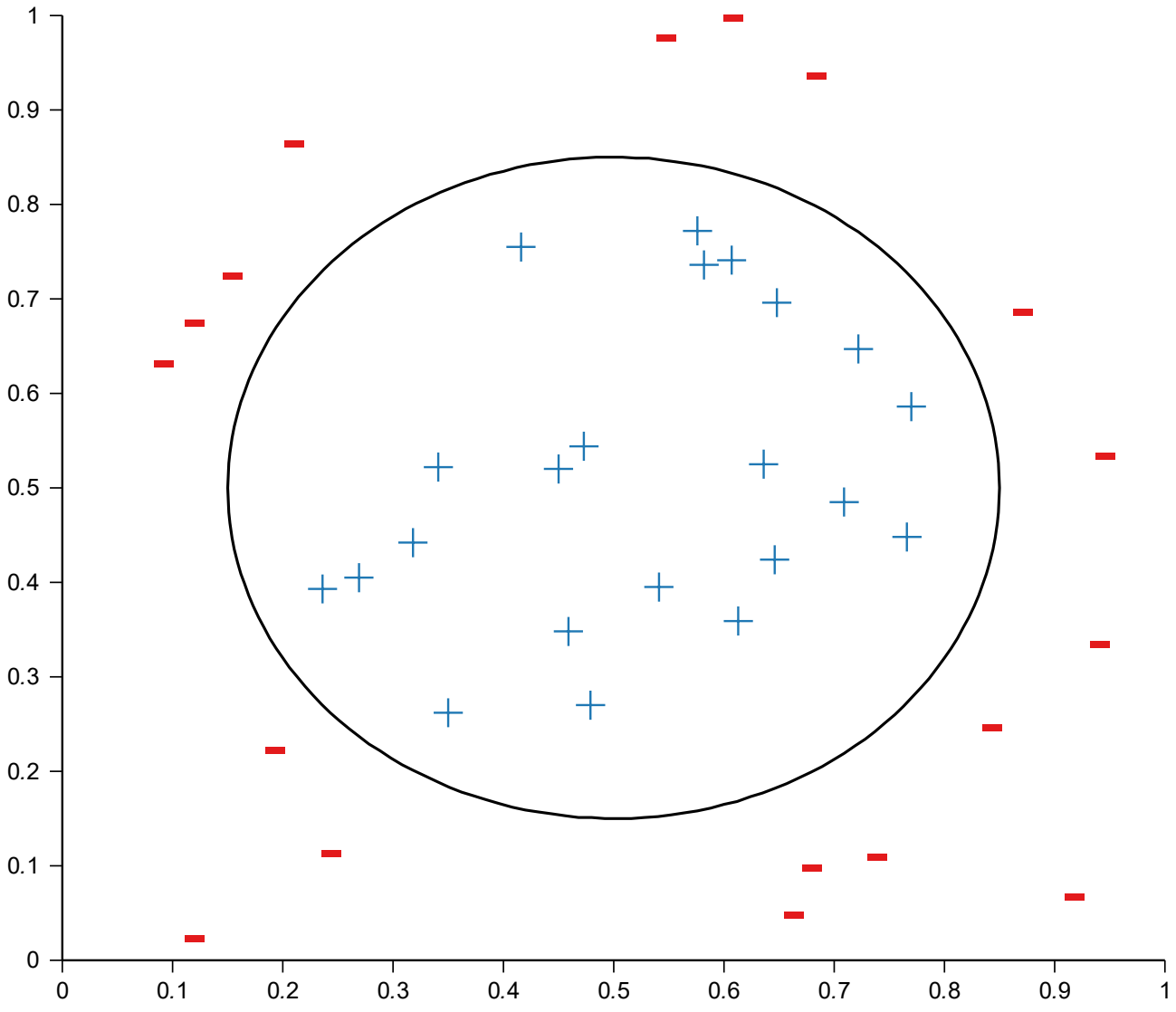


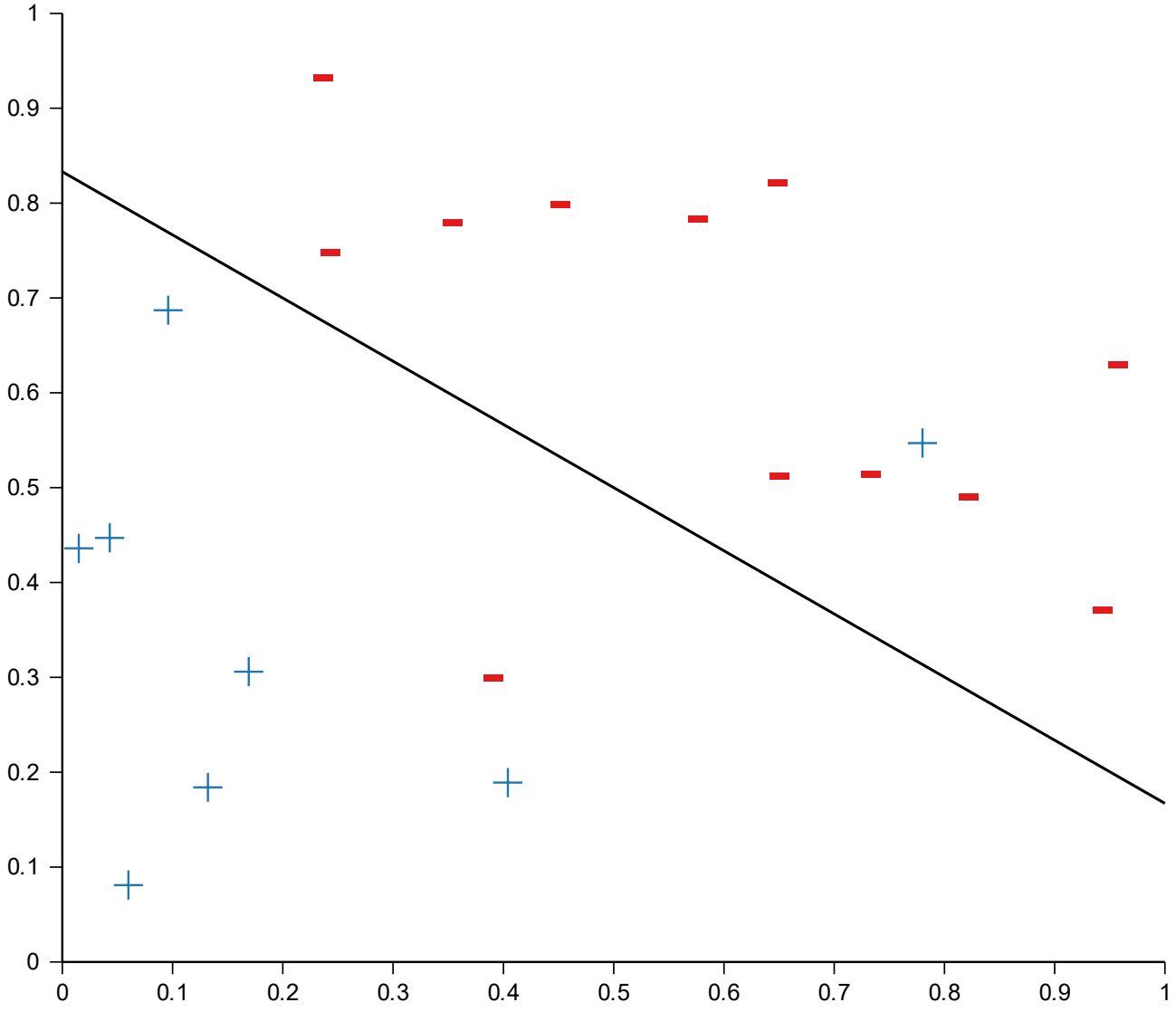
Linear Models?



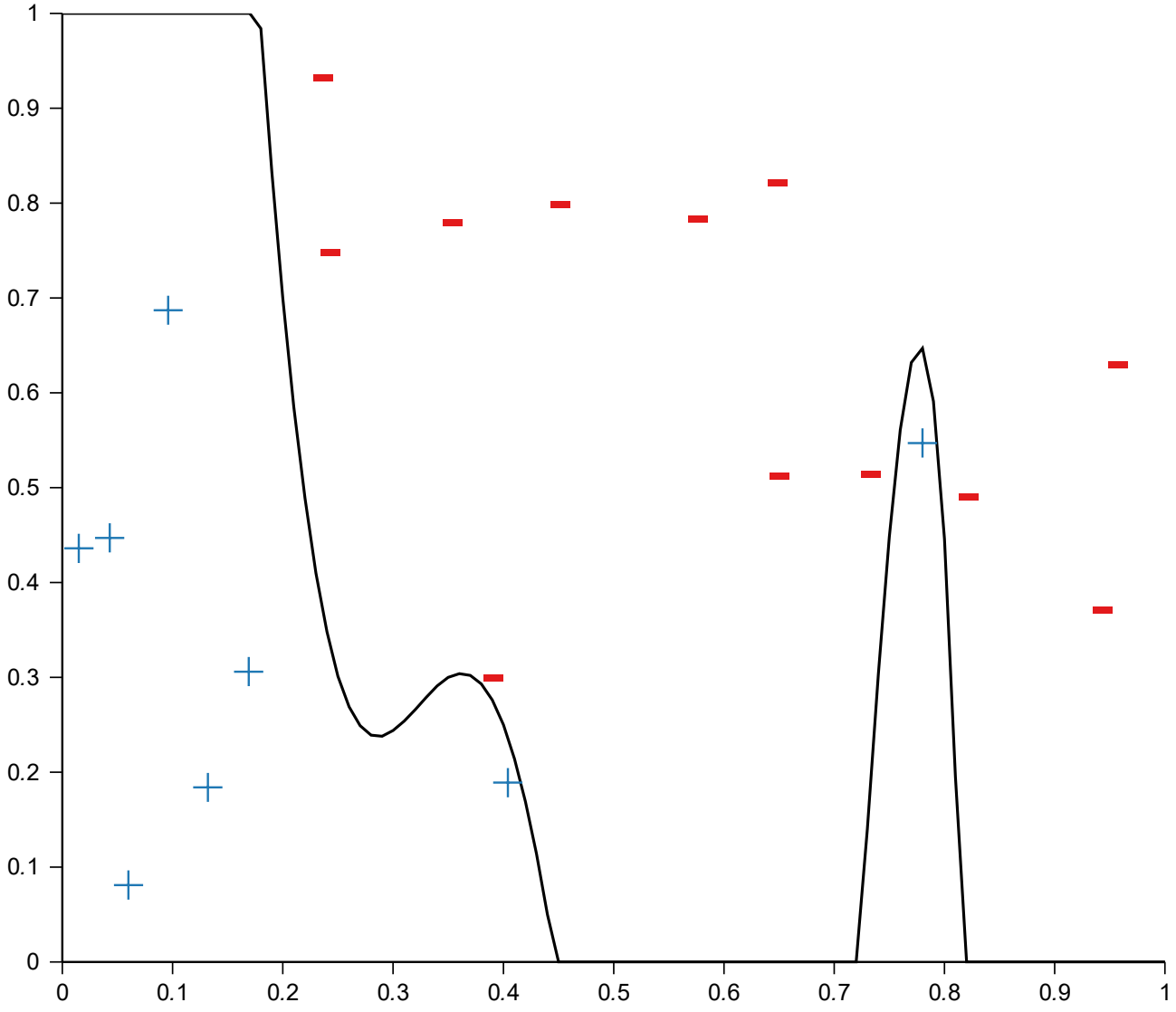
Nonlinear Models



Linear Models



Nonlinear Models?



Feature Transforms

- Given D-dimensional inputs $\boldsymbol{x} = [x_1, \dots, x_D]$, first compute some transformation of our input, e.g.,

$$\phi([x_1, x_2]) = [z_1 = (x_1 - 0.5)^2, z_2 = (x_2 - 0.5)^2]$$

General Q^{th}-order Transforms

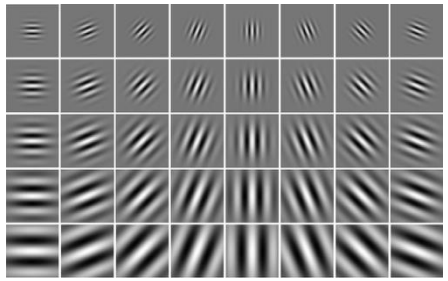
- $\phi_{2,2}([x_1, x_2]) = [x_1, x_2, x_1^2, x_1x_2, x_2^2]$
- $\phi_{2,3}([x_1, x_2]) = [x_1, x_2, x_1^2, x_1x_2, x_2^2, x_1^3, x_1^2x_2, x_1x_2^2, x_2^3]$
- $\phi_{2,4}([x_1, x_2]) =$
  $[x_1, x_2, x_1^2, x_1x_2, x_2^2, x_1^3, x_1^2x_2, x_1x_2^2, x_2^3, x_1^4, x_1^3x_2, x_1^2x_2^2, x_1x_2^3, x_2^4]$
- $\phi_{2,Q}$ maps a 2-D input to a $O(Q^2)$-D output
- Scales even worse for higher-dimensional inputs…

Deterministic Transforms

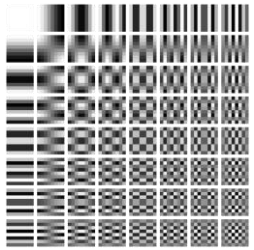
- log-transforms: x -> log(a + x)
- power/root transforms: x -> x^a
- polynomial transforms: x -> (x, x^2, x^3,…)

Harmonic Transforms

- Gabor Filters:



- Discrete Cosine Transforms:



- …

Can also have Linear Transforms

$$X \mapsto \{w_j^T X\}_{j=1}^k$$

- Principal Component Analysis
- Independent Component Analysis

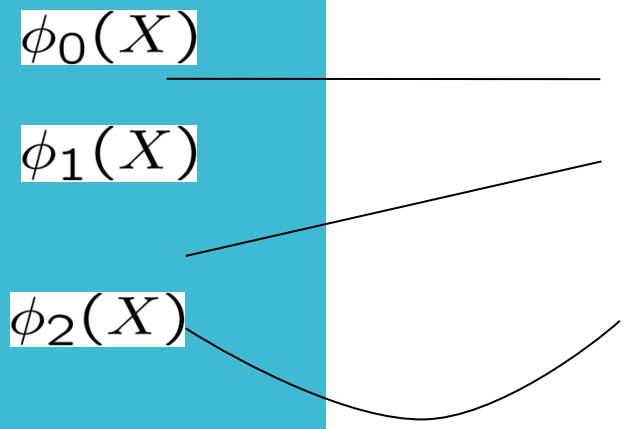
Regression with basis functions

$$f(X) = \sum_{j=0}^{m} \beta_j \phi_j(X)$$

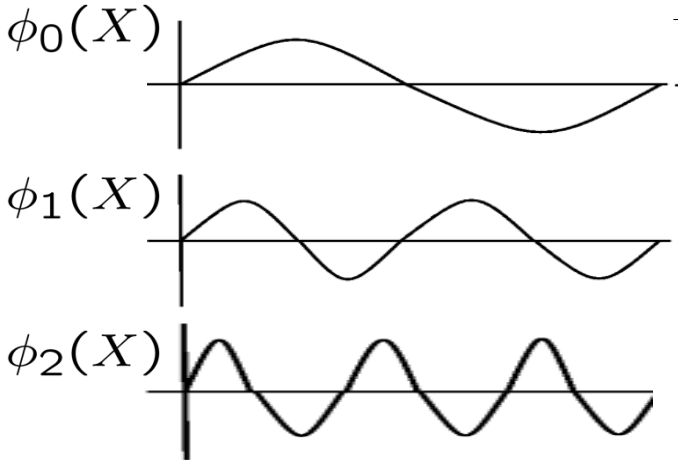
Basis coefficients

Basis functions (Linear combinations yield meaningful spaces)

Polynomial Basis

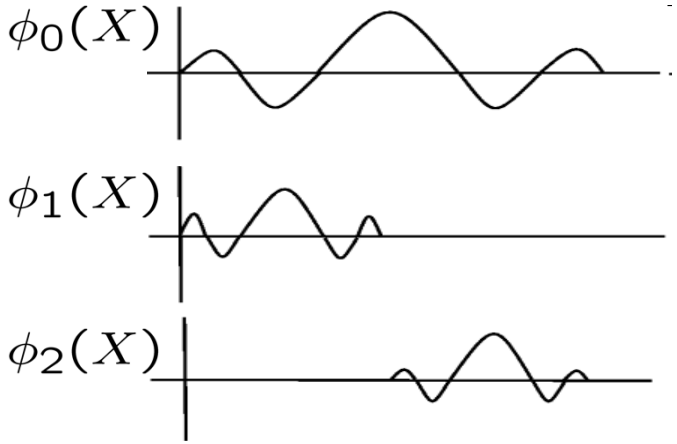


Fourier Basis



Good representation for periodic functions

Wavelet Basis

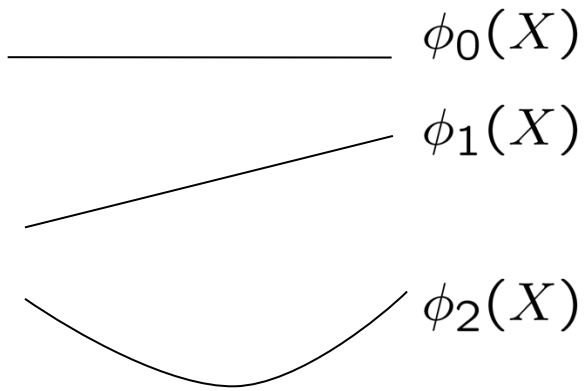


Good representation for local functions

Regression with nonlinear features

$$f(X) = \sum_{j=0}^{m} \beta_j X^j = \sum_{j=0}^{m} \beta_j \underbrace{\phi_j(X)}_{\text{Nonlinear features}}$$

Weight of each feature



In general, use any nonlinear features

e.g. e^X, log X, 1/X, sin(X), …

$$\widehat{\beta} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y}$$

or

$$(\mathbf{A}^T\mathbf{A} + \lambda\mathbf{I})^{-1}\mathbf{A}^T\mathbf{Y}$$

$$\mathbf{A} = \begin{bmatrix} \phi_0(X_1) & \phi_1(X_1) & \dots & \phi_m(X_1) \\ \vdots & & \ddots & \vdots \\ \phi_0(X_n) & \phi_1(X_n) & \dots & \phi_m(X_n) \end{bmatrix}$$

$$\widehat{f}_n(X) = \mathbf{X}\widehat{\beta}$$

$$\mathbf{X} = [\phi_0(X) \ \phi_1(X) \ \dots \ \phi_m(X)]$$

Polynomial Regression

Univariate (1-dim) case:

degree m

$$f(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_m X^m = \mathbf{X}\beta$$

where $\mathbf{X} = [1 \;\; X \;\; X^2 \ldots X^m] \quad \beta = [\beta_1 \ldots \beta_m]^T$

$$\widehat{\beta} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y} \quad \text{or} \quad (\mathbf{A}^T\mathbf{A} + \lambda\mathbf{I})^{-1}\mathbf{A}^T\mathbf{Y} \qquad \widehat{f}_n(X) = \mathbf{X}\widehat{\beta}$$

where $\mathbf{A} = \begin{bmatrix} 1 & X_1 & X_1^2 & \ldots & X_1^m \\ \vdots & & & \ddots & \vdots \\ 1 & X_n & X_n^2 & \ldots & X_n^m \end{bmatrix}$

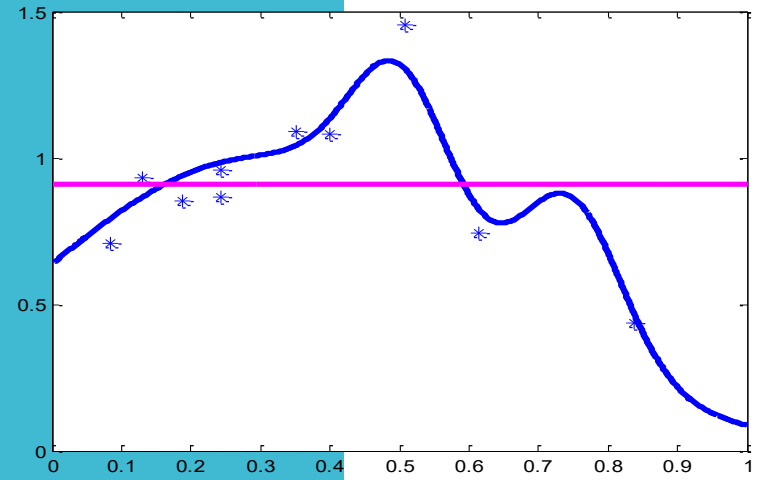
Multivariate (p-dim) case:

$$\begin{aligned} f(X) \quad = \quad & \beta_0 + \beta_1 X^{(1)} + \beta_2 X^{(2)} + \cdots + \beta_p X^{(p)} \\ & + \sum_{i=1}^{p}\sum_{j=1}^{p} \beta_{ij} X^{(i)} X^{(j)} + \sum_{i=1}^{p}\sum_{j=1}^{p}\sum_{k=1}^{p} X^{(i)} X^{(j)} X^{(k)} \\ & + \ldots \text{terms up to degree m} \end{aligned}$$

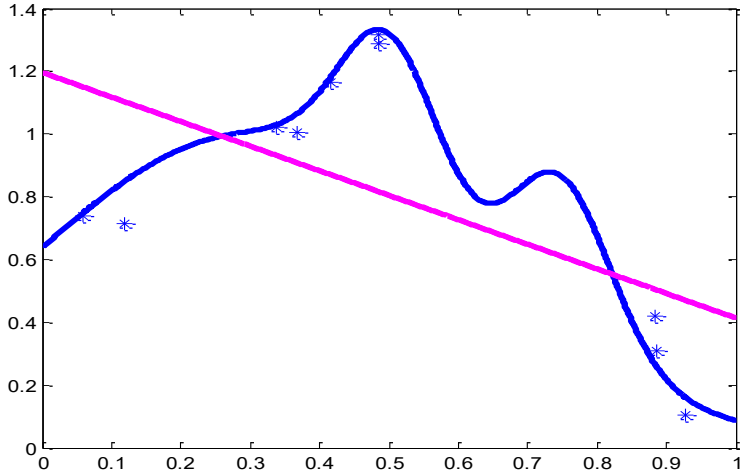
Polynomial Regression

Polynomial of order k, equivalently of degree up to k-1

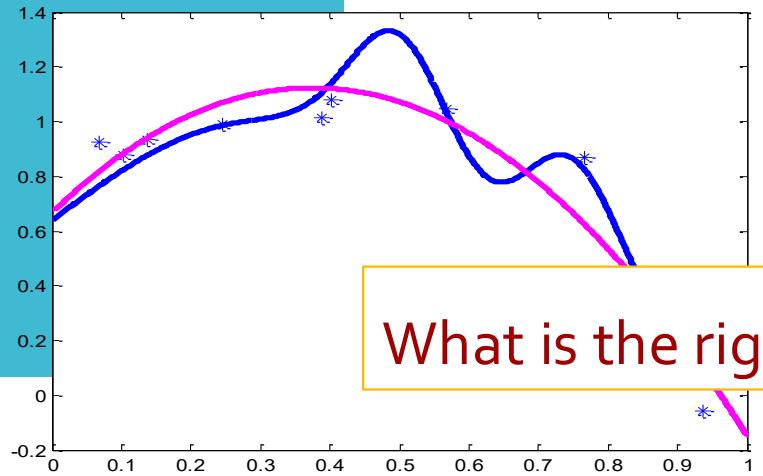
k=1



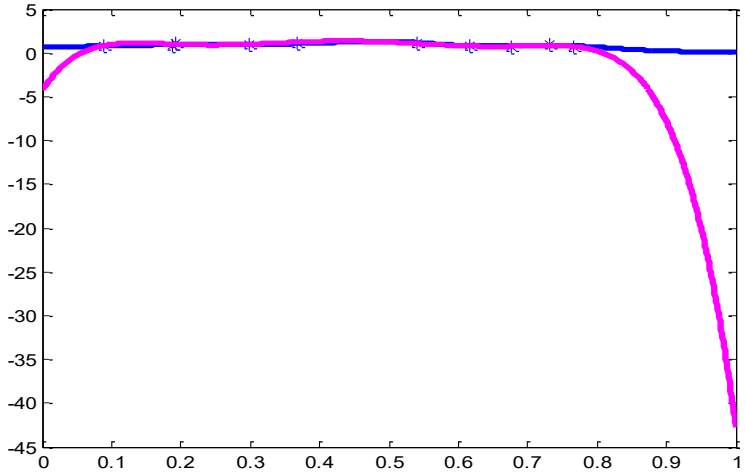
k=2



k=3



k=7

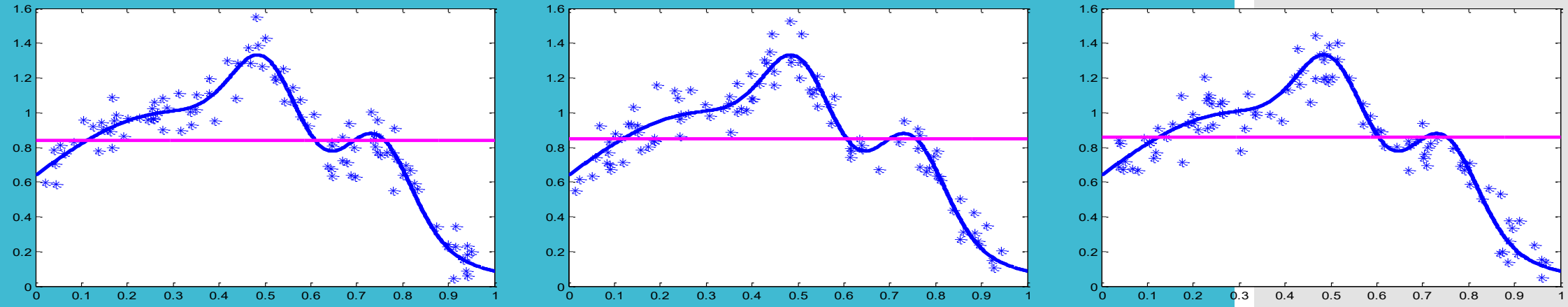


What is the right order?

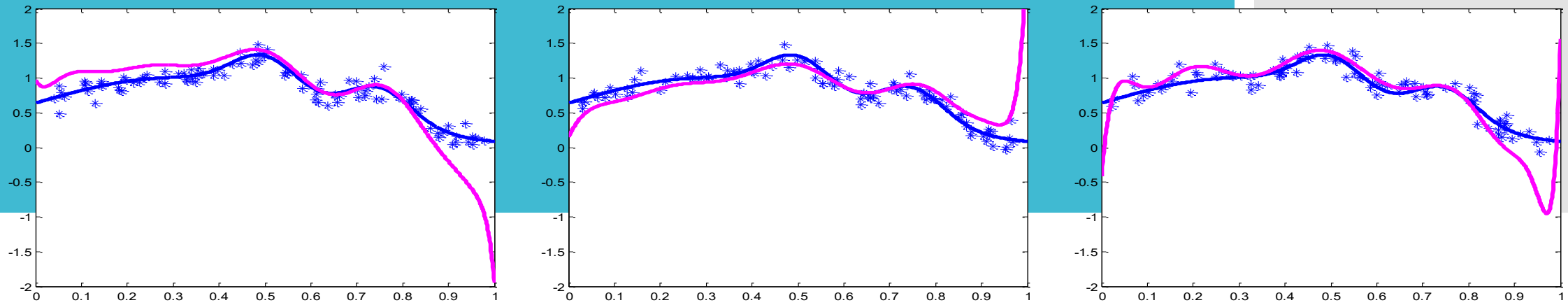
Bias – Variance Tradeoff

3 Independent training datasets

Large bias, Small variance – poor approximation but robust/stable



Small bias, Large variance – good approximation but unstable



Bias – Variance Decomposition

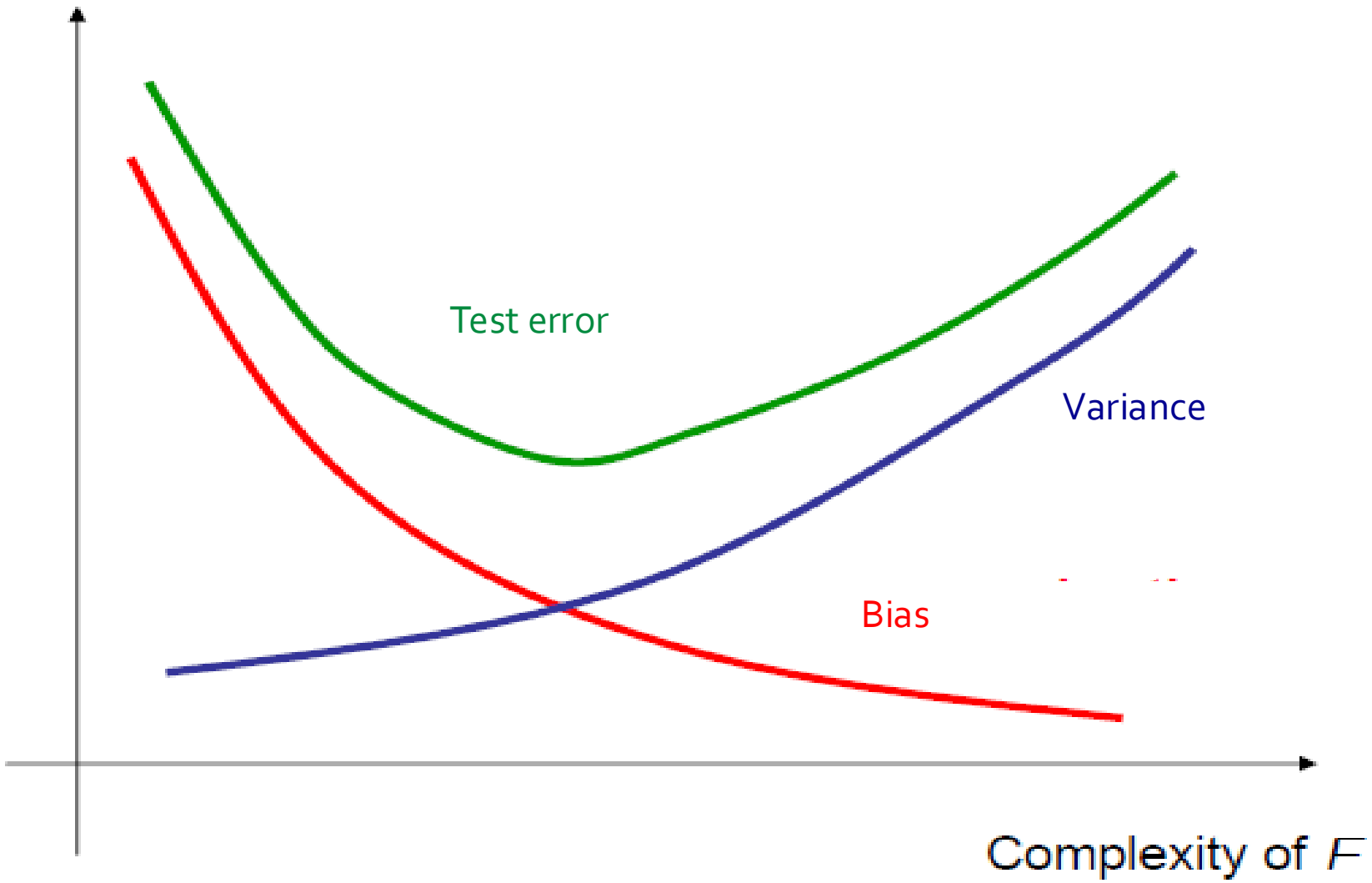
Later in the course, we will show that

$E[(f(X) - f^*(X))^2] = \text{Bias}^2 + \text{Variance}$

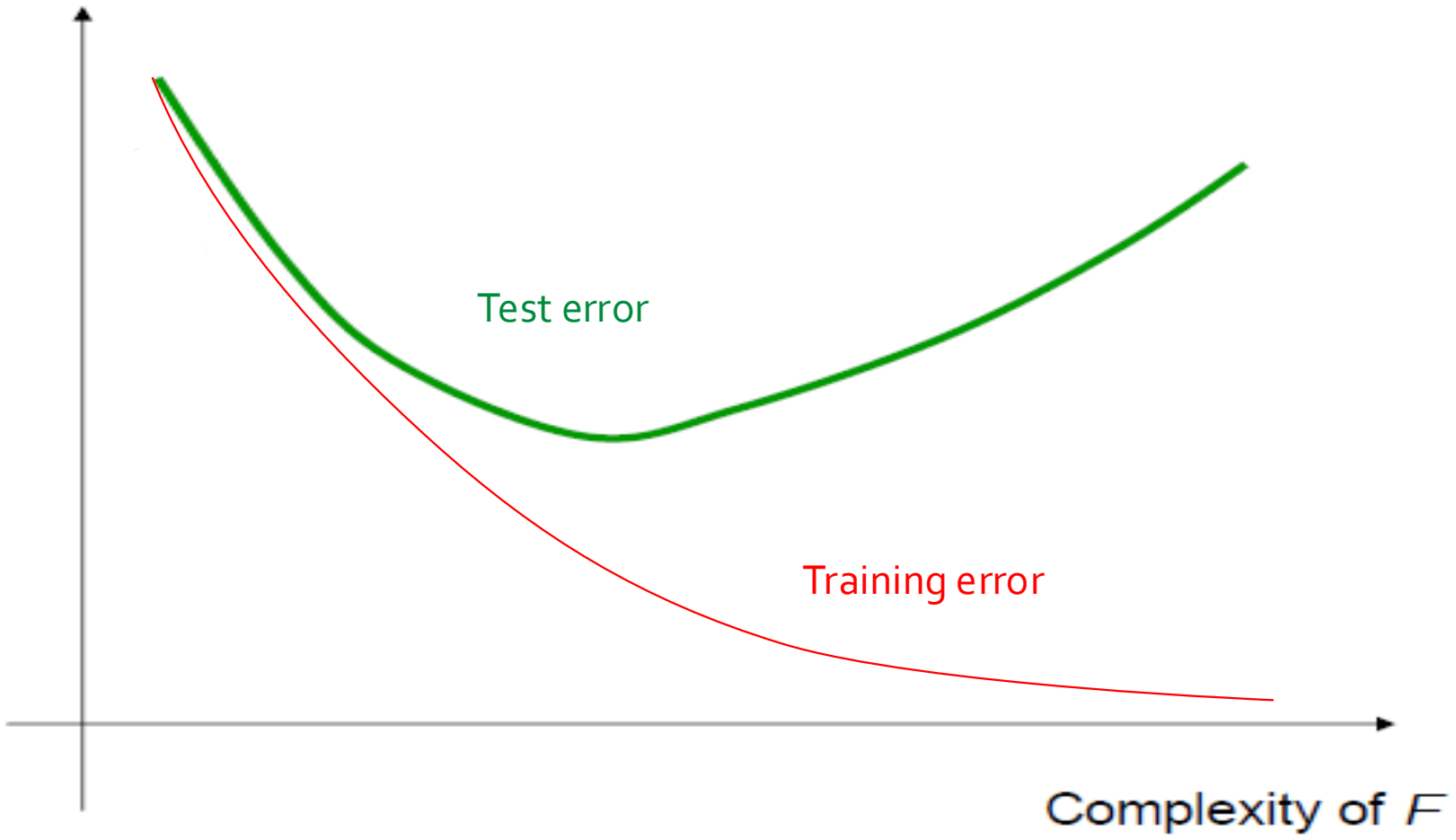
$\text{Bias} = E[f(X)] - f^*(X)$ ..... How far is the model from "true function"

$\text{Variance} = E[(f(X) - E[f(X)])^2]$ ..... How variable/stable is the model

Effect of Model Complexity



Effect of Model Complexity



Efficiency

- Depending on the transformation Φ and the dimensionality of the original input space $\mathcal{X}$, computing $\Phi(\boldsymbol{x})$ can be prohibitively computationally expensive
  - Computing $\Phi_2(\boldsymbol{x}) = [x_1, x_2, \dots, x_D, x_1^2, x_1 x_2, \dots, x_D^2]$ for $x \in \mathbb{R}^D$ requires $D + \binom{D}{2} + D = \frac{D^2+3D}{2} = O(D^2)$ time
  - Computing $\Phi_{10}(\boldsymbol{x})$ requires $O(D^{10})$ time
- Tradeoff:
  - High-dimensional transformations can result in good hypotheses (as long as they don't overfit) but…
  - High-dimensional transformations are expensive

Inner Product Methods

- Insight: the predictions of many machine learning models can be expressed as a function of inner products between feature vectors i.e., given $\mathcal{D} = \left\{\left(\boldsymbol{x}^{(n)}, y^{(n)}\right)\right\}_{n=1}^{N}$

$$h_{\mathcal{D}}(\boldsymbol{x}) = g\left(\left\{\boldsymbol{x}^T\boldsymbol{x}^{(n)}\right\}_{n=1}^{N}, \left\{\boldsymbol{x}^{(n)^T}\boldsymbol{x}^{(m)}\right\}_{n,m=1}^{N}\right)$$

- Crucially, feature vectors ***only*** appear in inner products with other feature vectors
- Applying a feature transformation Φ gives

$$h_{\mathcal{D}}\left(\Phi(\boldsymbol{x})\right) = g\left(\left\{\Phi(\boldsymbol{x})^T\Phi\left(\boldsymbol{x}^{(n)}\right)\right\}_{n=1}^{N}, \left\{\Phi\left(\boldsymbol{x}^{(n)}\right)^T\Phi\left(\boldsymbol{x}^{(m)}\right)\right\}_{n,m=1}^{N}\right)$$

The Kernel Trick

- Idea: for inner product methods, instead of computing $\Phi(\boldsymbol{x})$, use some function K_Φ s.t. $K_\Phi(\boldsymbol{x}, \boldsymbol{x}') = \Phi(\boldsymbol{x})^T \Phi(\boldsymbol{x}') \; \forall \, \boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}$
  - $K_\Phi(\boldsymbol{x}, \boldsymbol{x}')$ should be cheaper to compute than $\Phi(\boldsymbol{x})$
- Example: $\Phi_2'(\boldsymbol{x}) = \left[x_1, \dots, x_D, x_1^2, \sqrt{2}x_1x_2, \dots, \sqrt{2}x_{D-1}x_D, x_D^2\right]$

$$\Phi_2'(\boldsymbol{x})^T \Phi_2'(\boldsymbol{x}') = \sum_{i=1}^{D} x_i x_i' + \sum_{i=1}^{D} x_i^2 x_i'^2 + \sum_{i=1}^{D} \sum_{j \neq i} 2 x_i x_i' x_j x_j'$$

$$= \sum_{i=1}^{D} x_i x_i' + \left(\sum_{i=1}^{D} x_i x_i' \right)^2 = \boldsymbol{x}^T \boldsymbol{x}' + (\boldsymbol{x}^T \boldsymbol{x}')^2$$

$$K_{\Phi_2'}(\boldsymbol{x}, \boldsymbol{x}') = \boldsymbol{x}^T \boldsymbol{x}' + (\boldsymbol{x}^T \boldsymbol{x}')^2$$

- Computing $\Phi_2'(\boldsymbol{x})^T \Phi_2'(\boldsymbol{x}')$ requires $O(D^2)$ time whereas computing $K_{\Phi_2'}(\boldsymbol{x}, \boldsymbol{x}')$ only takes $O(D)$!

Common Kernels

- $K_{\Phi_2'}(\boldsymbol{x}, \boldsymbol{x}') = \boldsymbol{x}^T\boldsymbol{x}' + (\boldsymbol{x}^T\boldsymbol{x}')^2$
  - Implied feature transformation:

$$\Phi_2'(\boldsymbol{x}) = \left[x_1, \dots, x_D, x_1^2, \sqrt{2}x_1x_2, \dots, \sqrt{2}x_{D-1}x_D, x_D^2\right]$$

  - Implied dimensionality: $\frac{D^2+3D}{2}$

- $K_{\Phi_2^{(\gamma)}}(\boldsymbol{x}, \boldsymbol{x}') = (1 + \gamma\boldsymbol{x}^T\boldsymbol{x}')^2 - 1$
  - Implied feature transformation:

$$\Phi_2^{(\gamma)}(\boldsymbol{x}) = \left[\sqrt{2\gamma}x_1, \dots, \sqrt{2\gamma}x_D, \gamma x_1^2, \gamma x_1x_2, \dots, \gamma x_D^2\right]$$

    - γ affects the geometry of the transform

  - Implied dimensionality: $\frac{D^2+3D}{2}$

Common Kernels

- Polynomial Kernel: $K_{\Phi_Q^{(\gamma)}}(\boldsymbol{x}, \boldsymbol{x}') = (1 + \gamma \boldsymbol{x}^T \boldsymbol{x}')^Q - 1$
  - Implied dimensionality: $O(D^Q)$
  - γ affects the geometry of the transform
- Gaussian-RBF Kernel: $K_{\Phi_r}(\boldsymbol{x}, \boldsymbol{x}') = e^{-\frac{\|\boldsymbol{x}-\boldsymbol{x}'\|^2}{2r}}$
  - Implied feature transformation: $\Phi_r(\boldsymbol{x}) = \left[e^{-\frac{x_1^2}{2r}}, \dots, e^{-\frac{x_D^2}{2r}}, \right.$

$$\left. e^{-\frac{x_1^2}{2r}} \sqrt{\frac{(x_1)^2}{1! r^1}}, \dots, e^{-\frac{x_D^2}{2r}} \sqrt{\frac{(x_D)^2}{1! r^1}}, e^{-\frac{x_1^2}{2r}} \sqrt{\frac{\left(x_1^2\right)^2}{2! r^2}}, \dots, e^{-\frac{x_D^2}{2r}} \sqrt{\frac{\left(x_D^2\right)^2}{2! r^2}}, \dots \right]$$

Common Kernels

- Polynomial Kernel: $K_{\Phi_Q^{(\gamma)}}(\boldsymbol{x}, \boldsymbol{x}') = (1 + \gamma \boldsymbol{x}^T \boldsymbol{x}')^Q - 1$
  - Implied dimensionality: $O(D^Q)$
  - γ affects the geometry of the transform
- Gaussian-RBF Kernel: $K_{\Phi_r}(\boldsymbol{x}, \boldsymbol{x}') = e^{-\frac{\|\boldsymbol{x}-\boldsymbol{x}'\|^2}{2r}}$
  - Implied feature transformation: $\Phi_r(x) =$

$$\left[\left[e^{-\frac{x_1^2}{2r}}\sqrt{\frac{\left(x_1^d\right)^2}{d!r^d}}, \dots, e^{-\frac{x_D^2}{2r}}\sqrt{\frac{\left(x_1^d\right)^2}{d!r^d}}\right] : d \in \mathbb{N}\right]$$

  - Implied dimensionality: ∞!

Kernels Everywhere!

- Any method that only depends on the Euclidean distance between data points is an inner product method:

$$\|\boldsymbol{x} - \boldsymbol{x}'\|_2 = \sqrt{(\boldsymbol{x} - \boldsymbol{x}')^T(\boldsymbol{x} - \boldsymbol{x}')} = \sqrt{\boldsymbol{x}^T\boldsymbol{x} - 2\boldsymbol{x}^T\boldsymbol{x}' + \boldsymbol{x}'^T\boldsymbol{x}'}$$

  - We can kernelize kNN!

Kernels Everywhere!

- We can also kernelize linear/ridge regression!

$$\widehat{\boldsymbol{\omega}}_{MAP} = (X^T X + \lambda_C I_{D+1})^{-1} X^T \boldsymbol{y}$$
$$= X^T (XX^T + \lambda_C I_N)^{-1} \boldsymbol{y}$$

$$XX^T = \begin{bmatrix} 1 & \boldsymbol{x}^{(1)^T} \\ 1 & \boldsymbol{x}^{(2)^T} \\ \vdots & \vdots \\ 1 & \boldsymbol{x}^{(N)^T} \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \boldsymbol{x}^{(1)} & \boldsymbol{x}^{(2)} & \cdots & \boldsymbol{x}^{(N)} \end{bmatrix}$$

Let $\boldsymbol{\alpha} = (XX^T + \lambda_C I_{D+1})^{-1} \boldsymbol{y} \rightarrow \widehat{\boldsymbol{\omega}}_{MAP} = X^T \boldsymbol{\alpha}$

$$\boldsymbol{\alpha} \in \mathbb{R}^N \rightarrow \widehat{\boldsymbol{\omega}}_{MAP} = \sum_{i=1}^{N} \alpha_i \begin{bmatrix} 1 \\ \boldsymbol{x}^{(i)} \end{bmatrix}$$

$$\rightarrow h(\boldsymbol{x}) = \widehat{\boldsymbol{\omega}}_{\boldsymbol{MAP}}^T \boldsymbol{x} = \sum_{i=1}^{N} \alpha_i \begin{bmatrix} 1 & \boldsymbol{x}^{(i)^T} \end{bmatrix} \begin{bmatrix} 1 \\ \boldsymbol{x} \end{bmatrix}$$

Valid Kernels

- Any function K is a valid kernel if and only if:
  - $\exists$ a transformation Φ s.t.

$$K(\boldsymbol{x}, \boldsymbol{x}') = \Phi(\boldsymbol{x})^T \Phi(\boldsymbol{x}') \ \forall \ \boldsymbol{x}, \boldsymbol{x}'$$

$$\Updownarrow$$

  - the Gram matrix

$$\mathrm{K} = \begin{bmatrix} K\left(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(1)}\right) & K\left(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}\right) & \cdots & K\left(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(N)}\right) \\ K\left(\boldsymbol{x}^{(2)}, \boldsymbol{x}^{(1)}\right) & K\left(\boldsymbol{x}^{(2)}, \boldsymbol{x}^{(2)}\right) & \cdots & K\left(\boldsymbol{x}^{(2)}, \boldsymbol{x}^{(N)}\right) \\ \vdots & \vdots & \ddots & \vdots \\ K\left(\boldsymbol{x}^{(N)}, \boldsymbol{x}^{(1)}\right) & K\left(\boldsymbol{x}^{(N)}, \boldsymbol{x}^{(2)}\right) & \cdots & K\left(\boldsymbol{x}^{(N)}, \boldsymbol{x}^{(N)}\right) \end{bmatrix}$$

    is symmetric and positive semi-definite $\forall$ sets $\left\{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(N)}\right\}$

Building New Kernels

- For any valid kernels K_1, K_2 with implied feature transformations Φ_1, Φ_2 and non-negative coefficients c_1, c_2, the following are all valid kernels:
  - $K(\boldsymbol{x}, \boldsymbol{x}') = c_1 K_1(\boldsymbol{x}, \boldsymbol{x}') + c_2 K_2(\boldsymbol{x}, \boldsymbol{x}')$

    $$\Phi(\boldsymbol{x}) = [\sqrt{c_1}\Phi_1(\boldsymbol{x}), \sqrt{c_1}\Phi_2(\boldsymbol{x})]$$
  - $K(\boldsymbol{x}, \boldsymbol{x}') = c_1 K_1(\boldsymbol{x}, \boldsymbol{x}') K_2(\boldsymbol{x}, \boldsymbol{x}')$

    $$\Phi(\boldsymbol{x}) = \left[\left\{\sqrt{c_1}\phi_i(\boldsymbol{x})\phi_j(\boldsymbol{x})\right\}_{\phi_i(\boldsymbol{x})\in\Phi_1(\boldsymbol{x}), \phi_j(\boldsymbol{x})\in\Phi_2(\boldsymbol{x})}\right]$$
  - $K(\boldsymbol{x}, \boldsymbol{x}') = e^{K_1(\boldsymbol{x}, \boldsymbol{x}')}$

Taylor series: $e^{K_1(\boldsymbol{x}, \boldsymbol{x}')} = 1 + K_1(\boldsymbol{x}, \boldsymbol{x}') + \frac{K_1(\boldsymbol{x}, \boldsymbol{x}')^2}{2!} + \frac{K_1(\boldsymbol{x}, \boldsymbol{x}')^3}{3!} + \cdots$

Key Takeaways

- Logistic regression
  - Logistic function induces a linear decision boundary
  - Conditional likelihood maximization
- Polynomial/non-linear feature transformations allow for learning non-linear functions/decision boundaries but can lead to overfitting…
  - Address with regularization (next class)!